IOT SERVERLESS FLEET MONITORING AND PREDICTIVE MAINTENANCE WITH AMAZON SAGEMAKER

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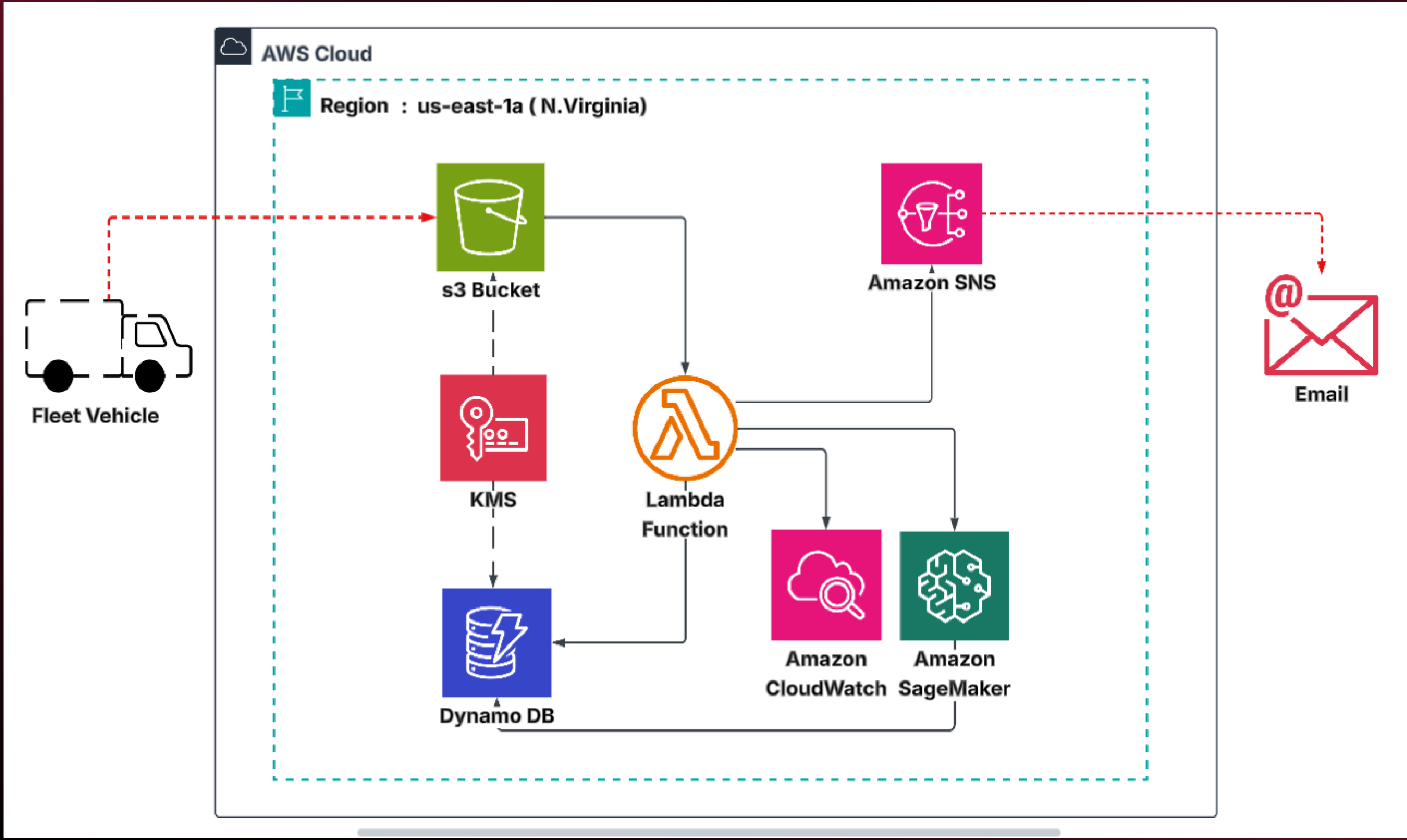
GitHub

PROJECT OVERVIEW

- This project focuses on tracking vehicle health and performance in real time using AWS serverless services.
- Fleet monitoring is continuously observing vehicles like trucks to detect issues such as high temperature, faults, or abnormal behavior.
- Predictive maintenance helps identify problems before they occur, reducing breakdowns and improving safety.
- In this system, vehicle temperature raw data is uploaded to S3, Lambda processes it, Amazon SageMaker predicts maintenance needs, DynamoDB stores results, and SNS sends instant alert emails to the driver/admin when critical issues are detected.

SERVICES USED

ARCHITECTURE



IAM ROLE

fleet-lambda-role > Create policy

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions f
role), attach a policy to it

Search

Allow (7 of 451 services)

Service	Access level	Resource
CloudWatch	Limited: Write	All resources
CloudWatch Logs	Limited: Write	Multiple
DynamoDB	Limited: Read, Write	region| string like |us-east-1, TableName| string like |fleet-table
KMS	Limited: Read, Write	region| string like |us-east-1, KeyID| string like |fleet-kms
S3	Limited: List, Read, Write	Multiple
SageMaker	Limited: Read	Multiple
SNS	Limited: Write	Multiple

fleet-sagemaker-role > Create policy

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for a
role), attach a policy to it

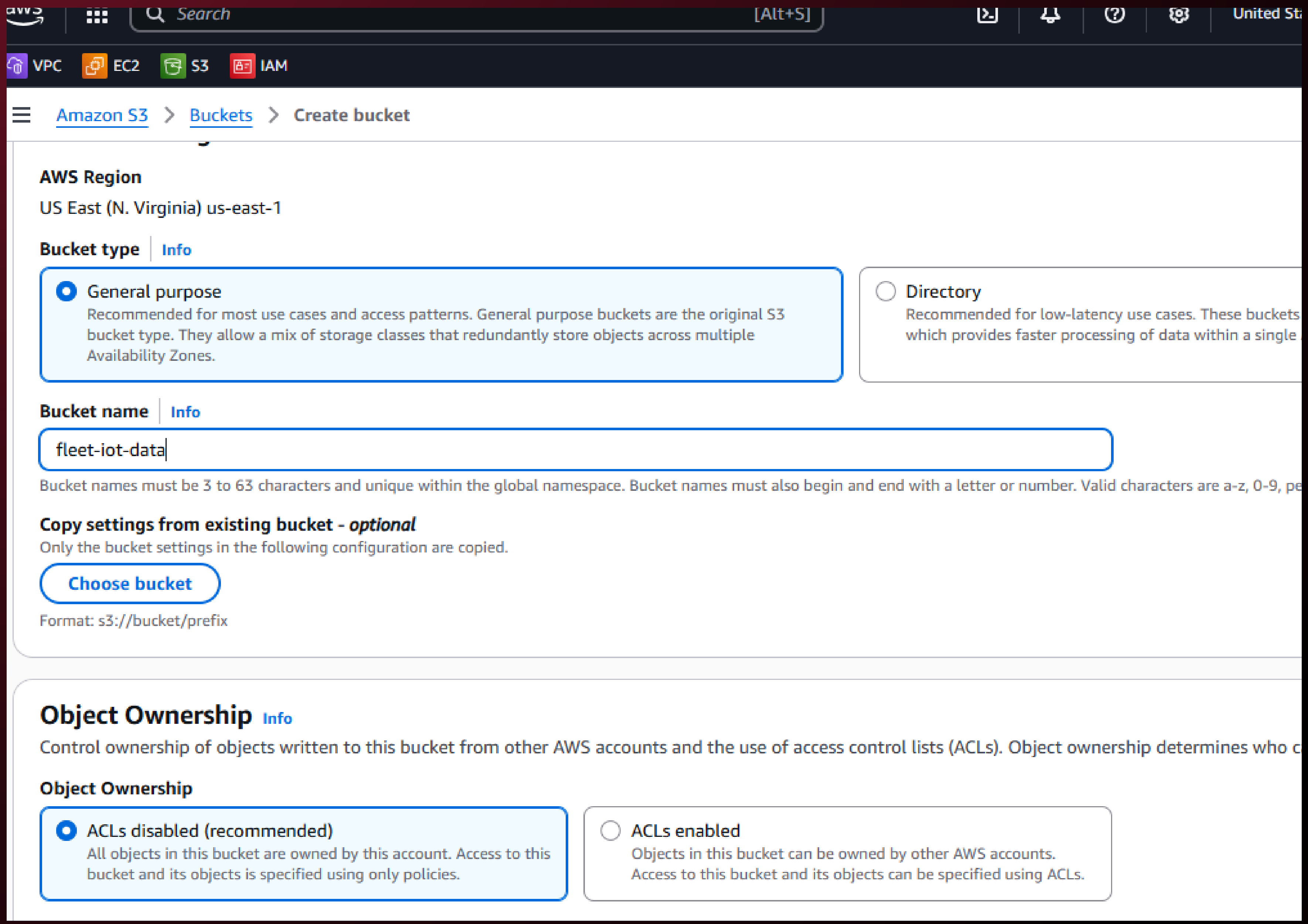
Search

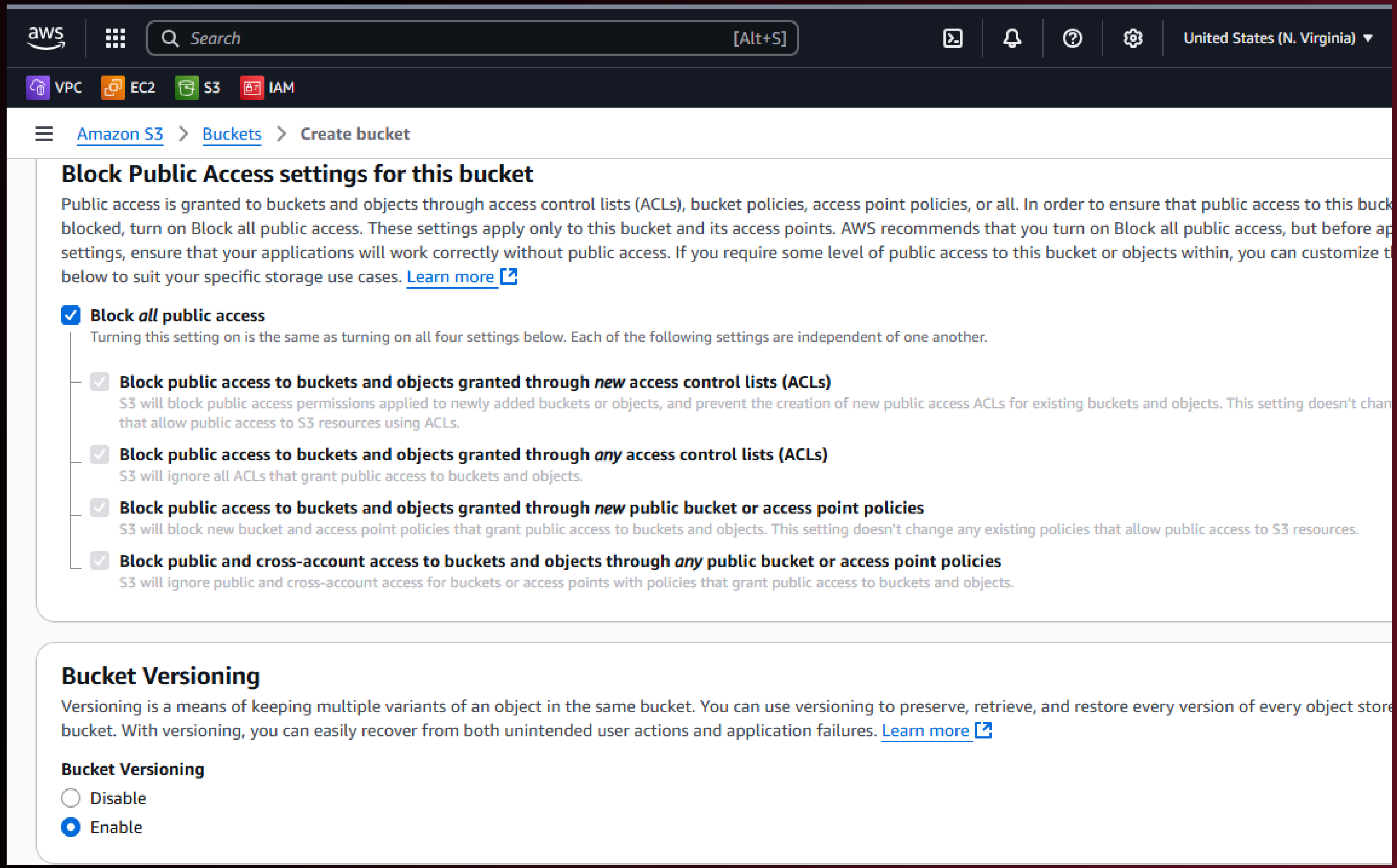
Allow (6 of 451 services)

Service	Access level	Resource
CloudWatch	Limited: Write	All resources
CloudWatch Logs	Limited: List, Read, Write	All resources
IAM	Limited: Write	RoleName| string like |fleet-sagemaker-role
KMS	Limited: Read, Write	region| string like |us-east-1, KeyID| string like |fleet-kms
S3	Limited: List, Permissions management, Read, Write, Tagging	Multiple
SageMaker	Full: Tagging Limited: List, Read, Write	All resources

Creating IAM role for Lambda and Sagemaker with permissions for S3,Sagemaker,KMS,Dynamo DB,Cloudwatch and SNS.

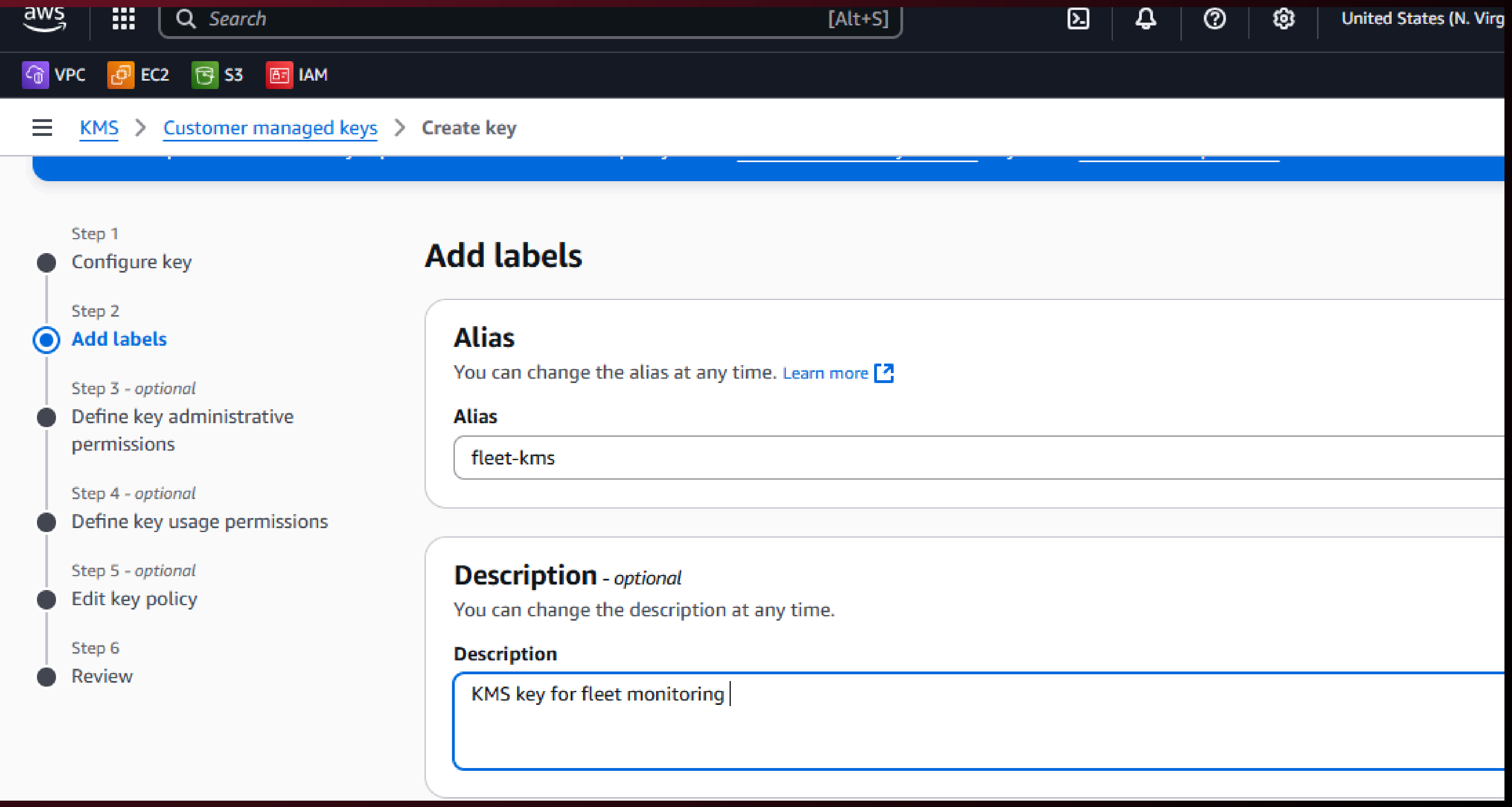
S3 BUCKET

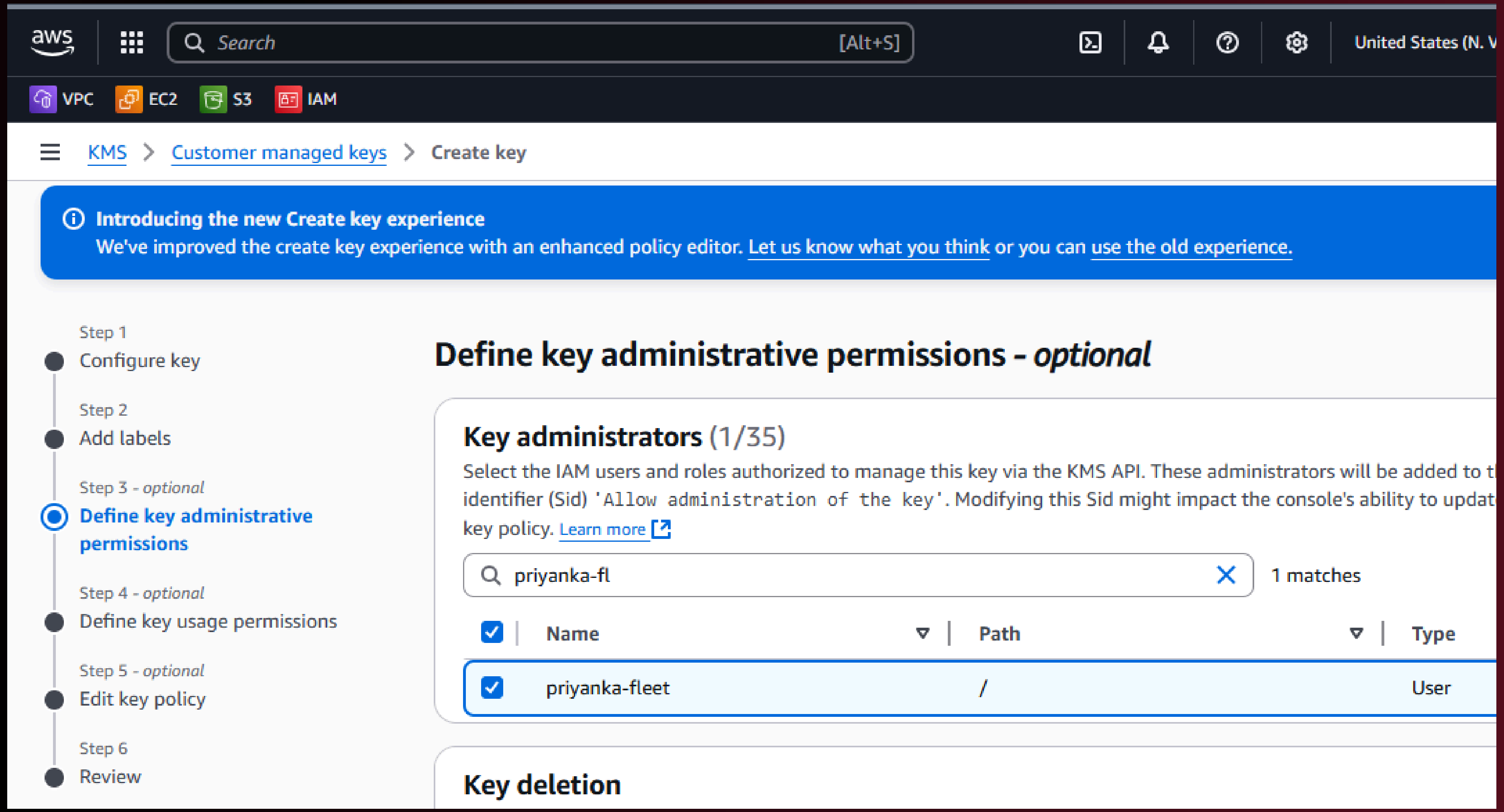


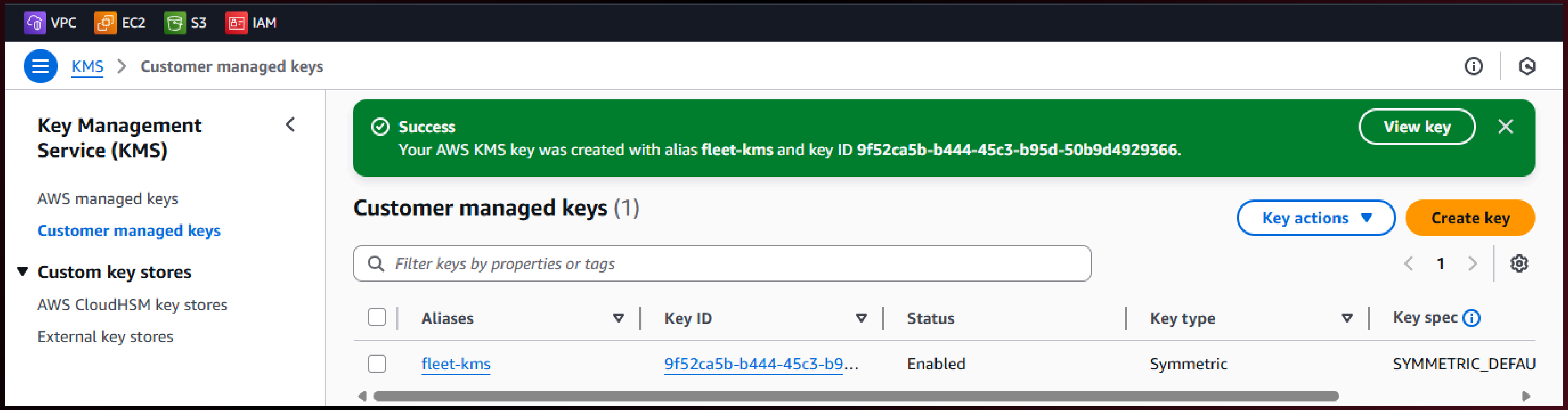


Creating S3 bucket with ACLs disabled ,public access blocked and bucket versioning enabled.

KEY MANAGEMENT SERVICE -KMS

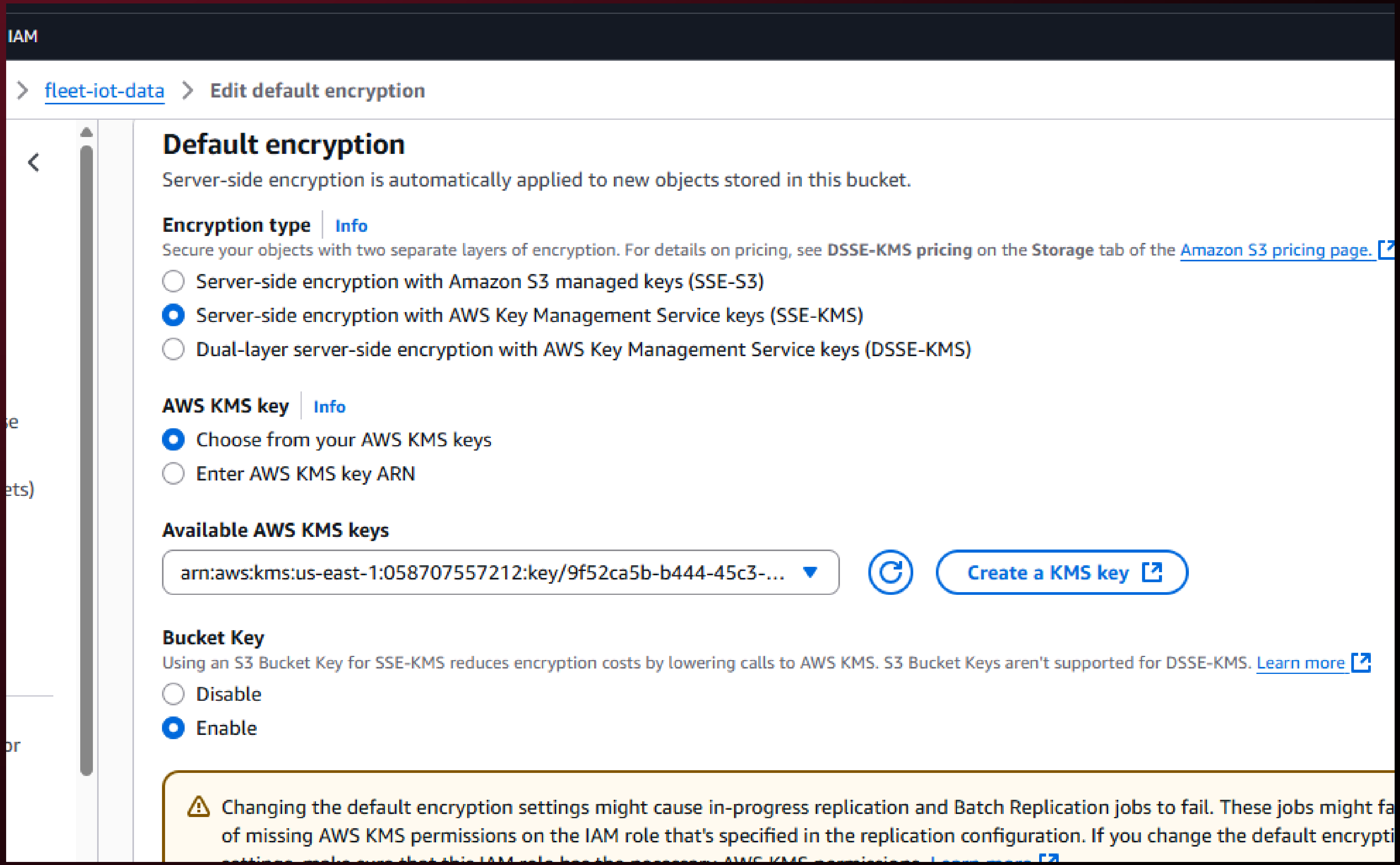


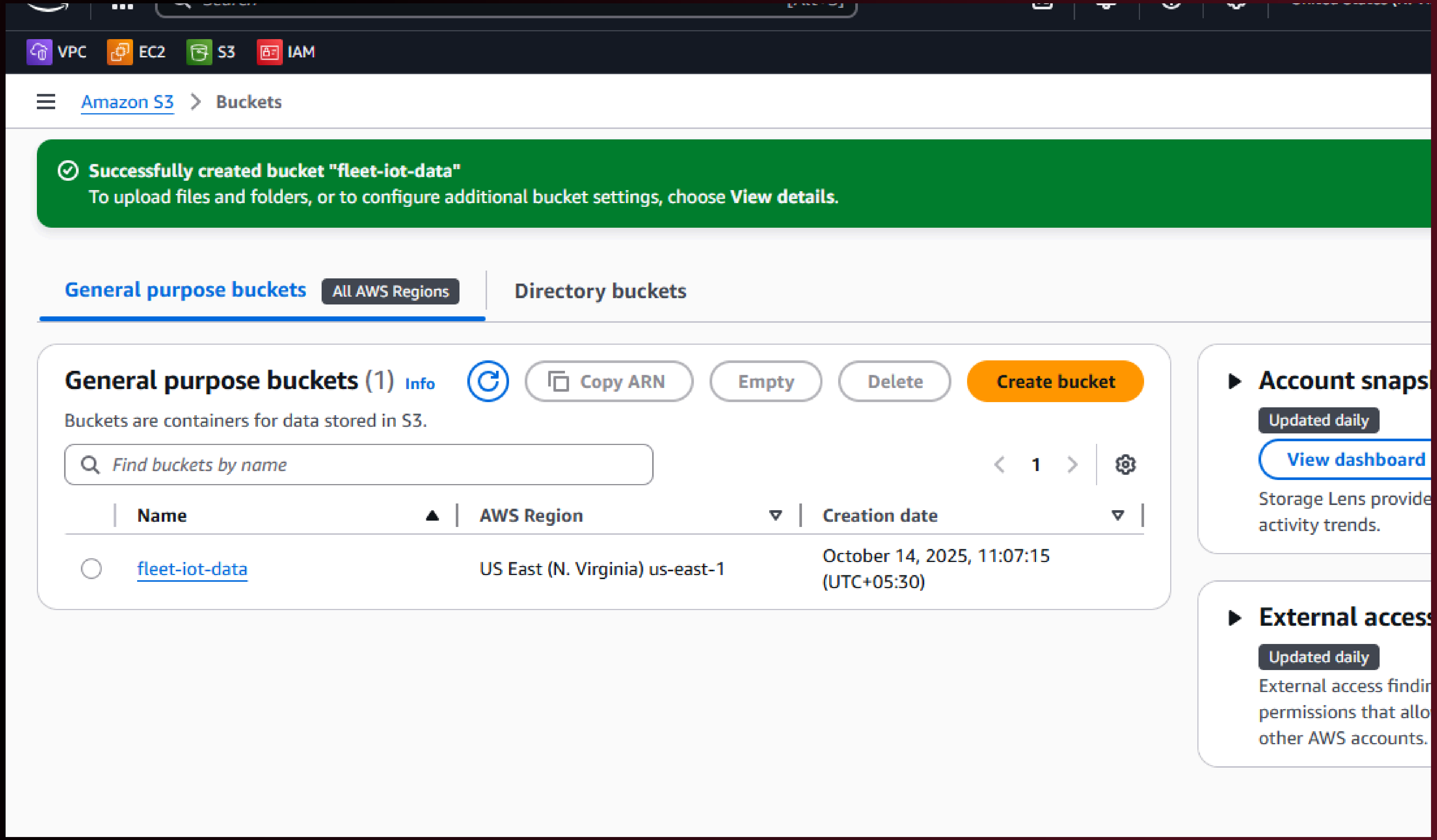




Creating KMS key with permisions given to respective IAM user to encrypt and decrypt S3 and Dynamo DB.Permission given to the respective IAM user ID.

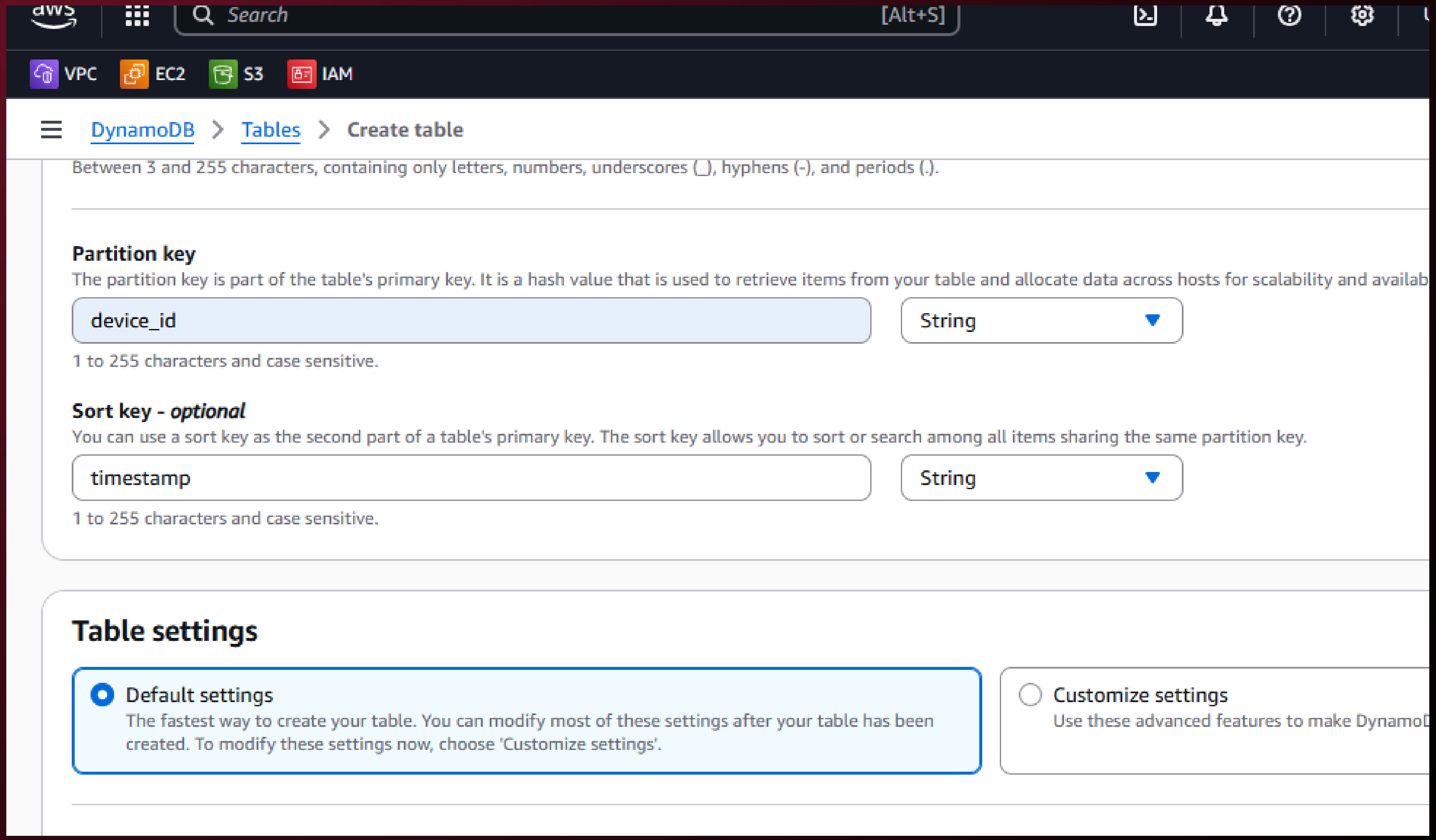
ATTACHING KMS KEY TO BUCKET

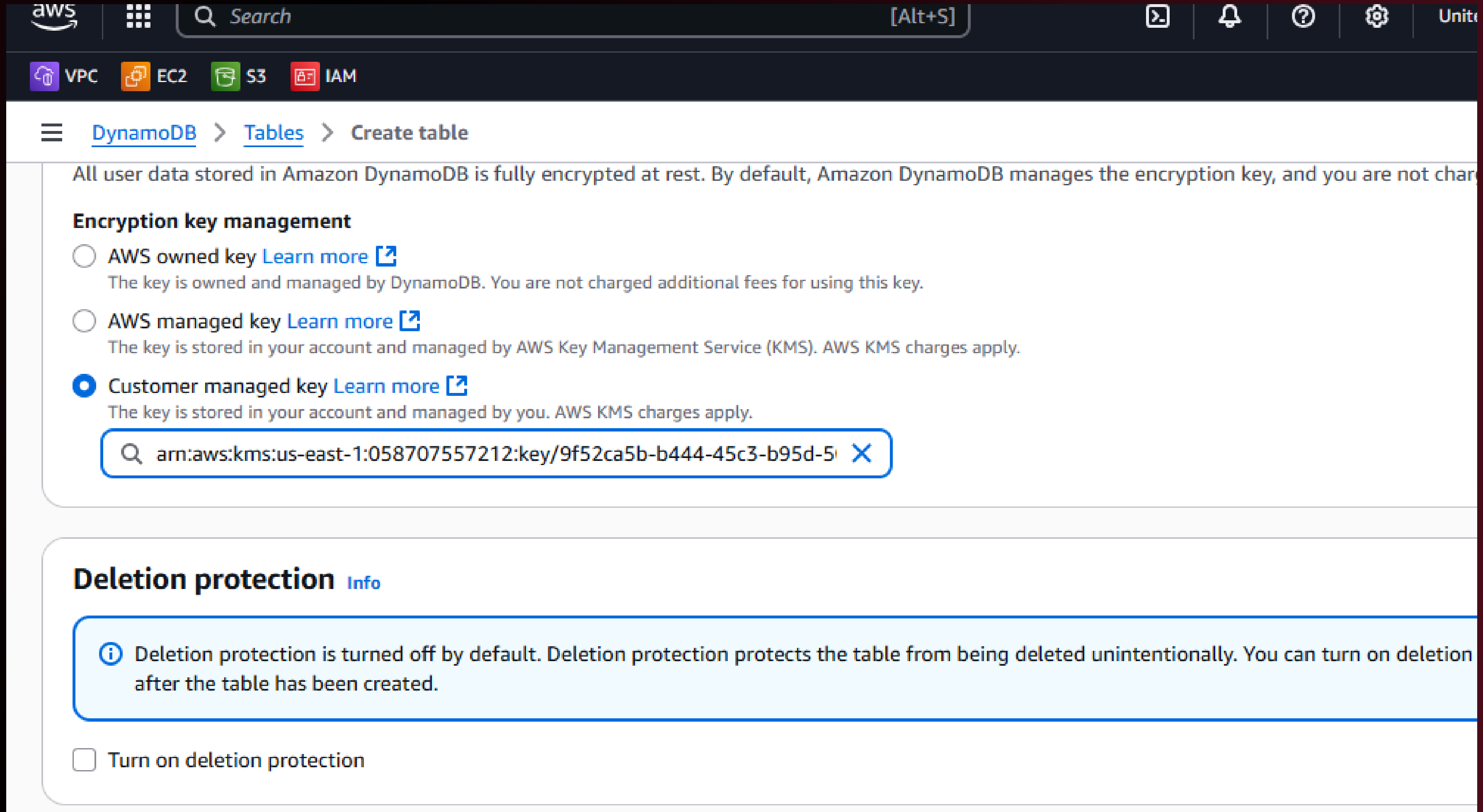


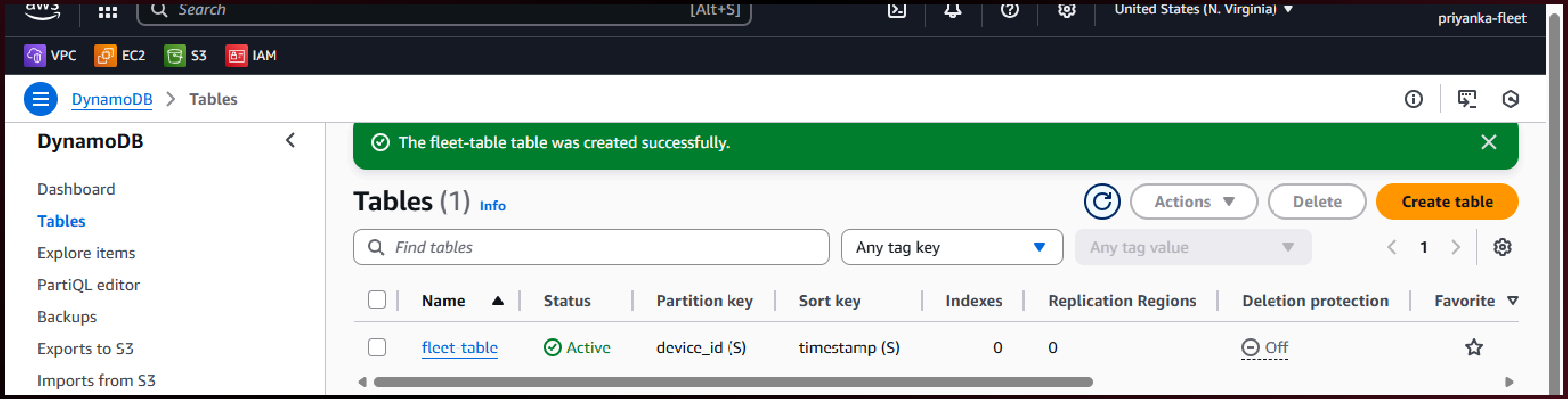


Attaching the KMS key to S3 bucket created.

DYNAMODB TABLE

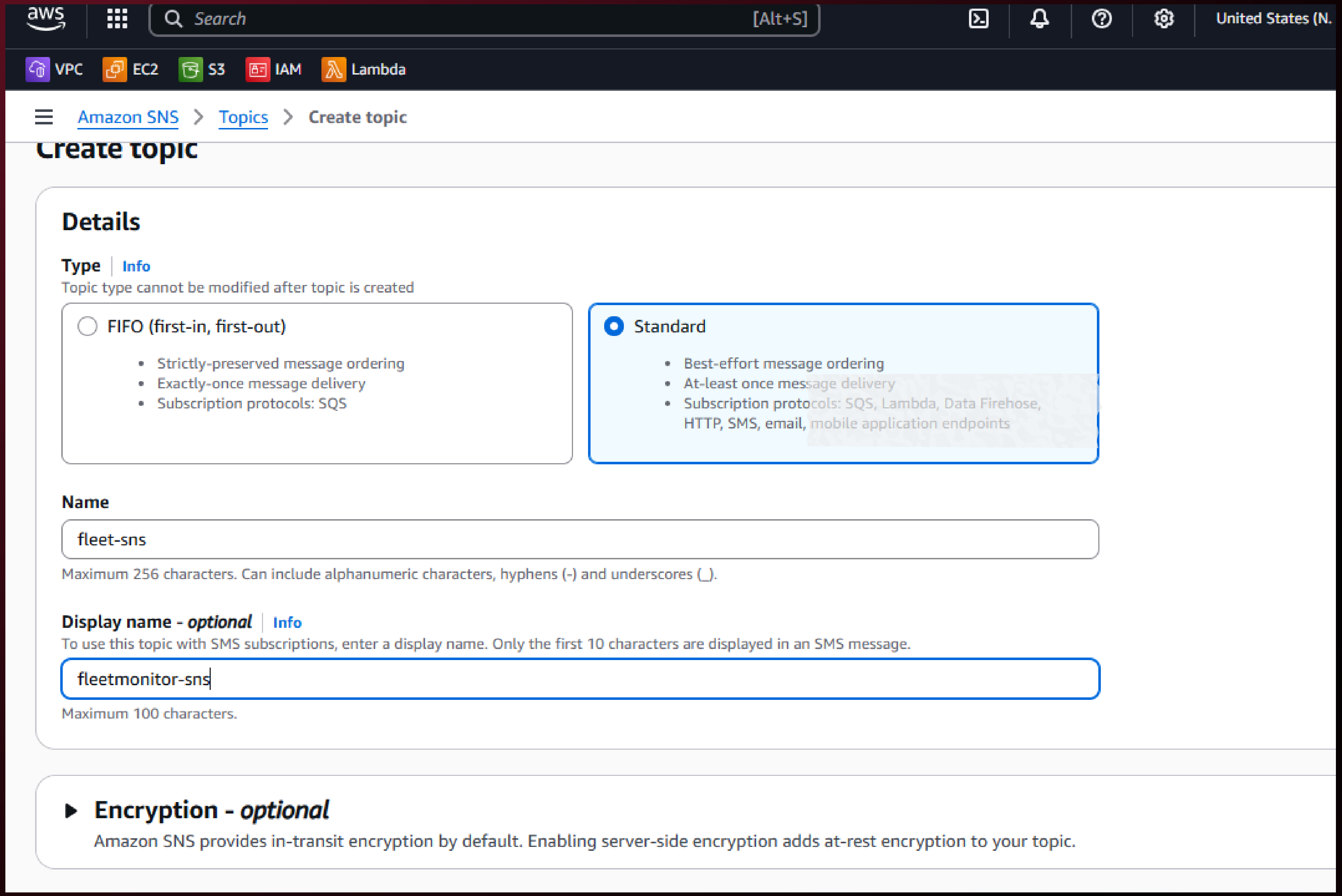


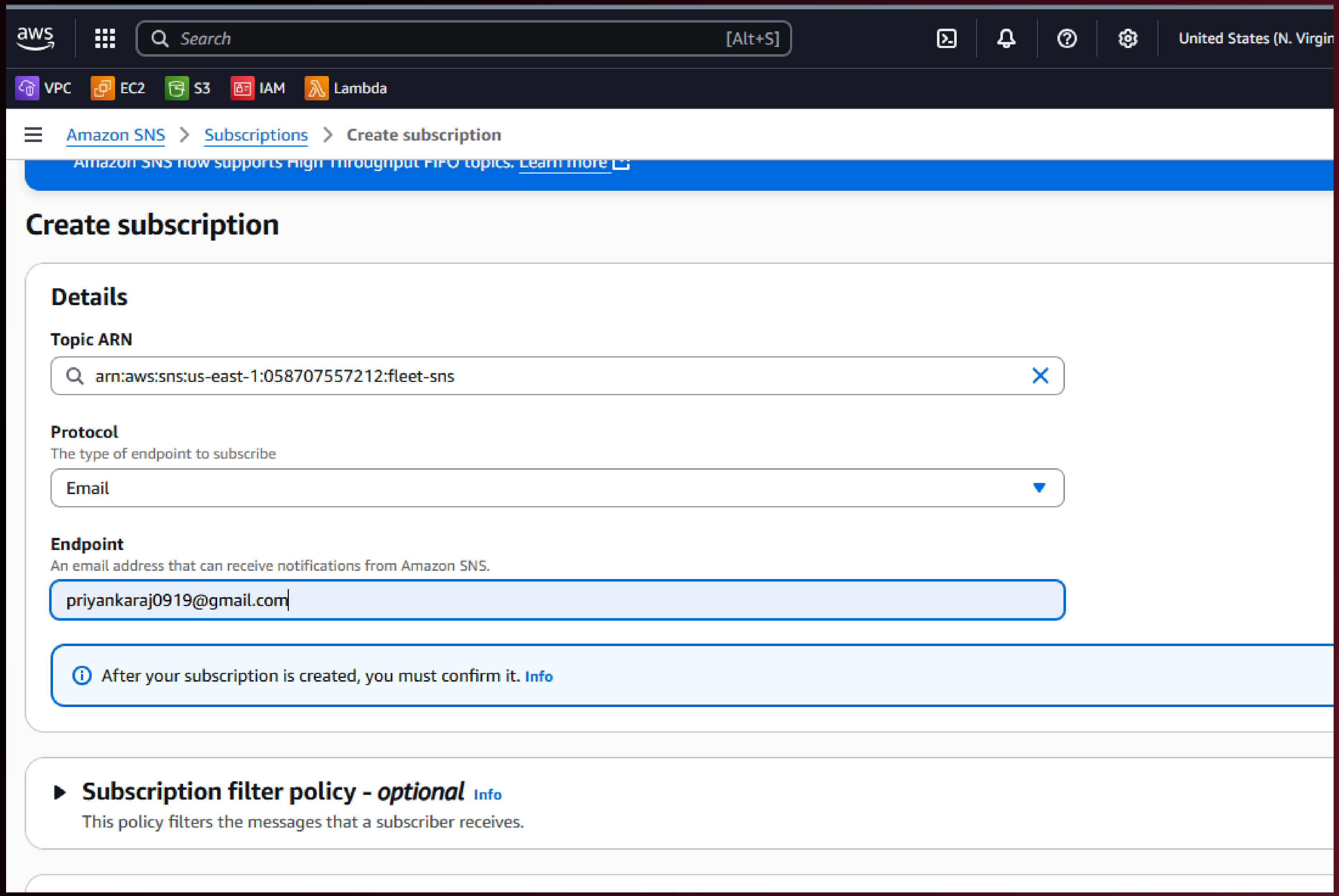




Created Dynamo DB table with the required partition key and Source key ,added the KMS key as well for encryption and decryption.

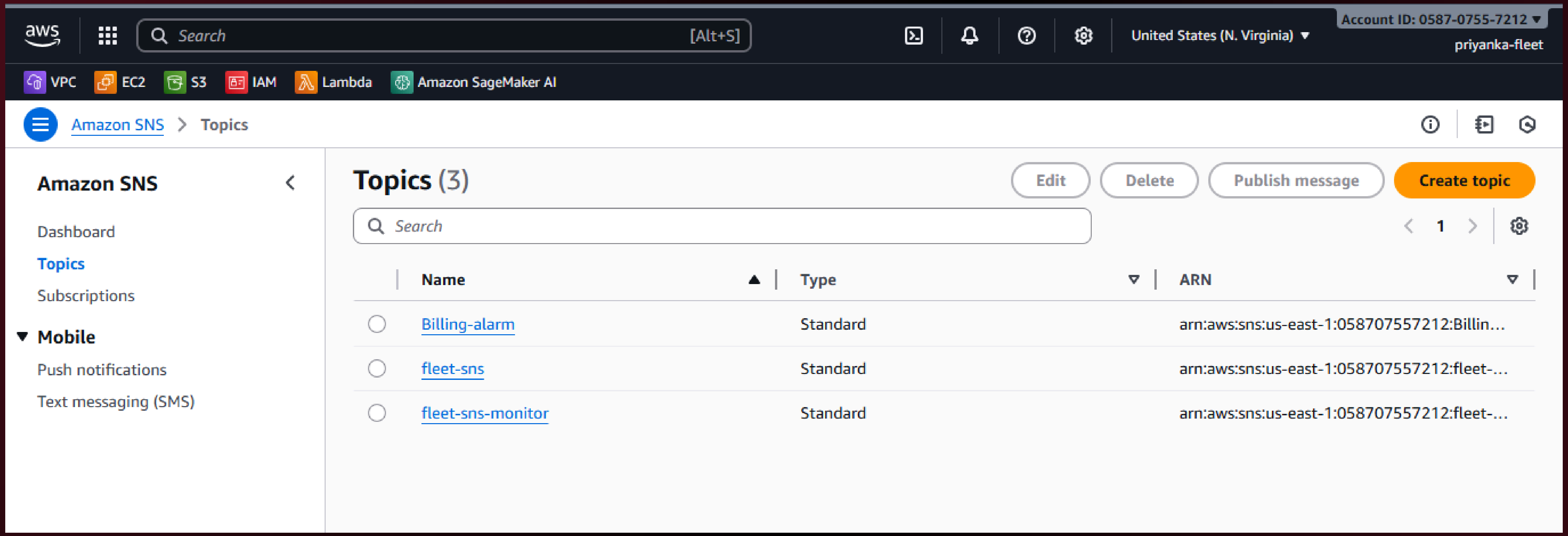
AMAZON SNS

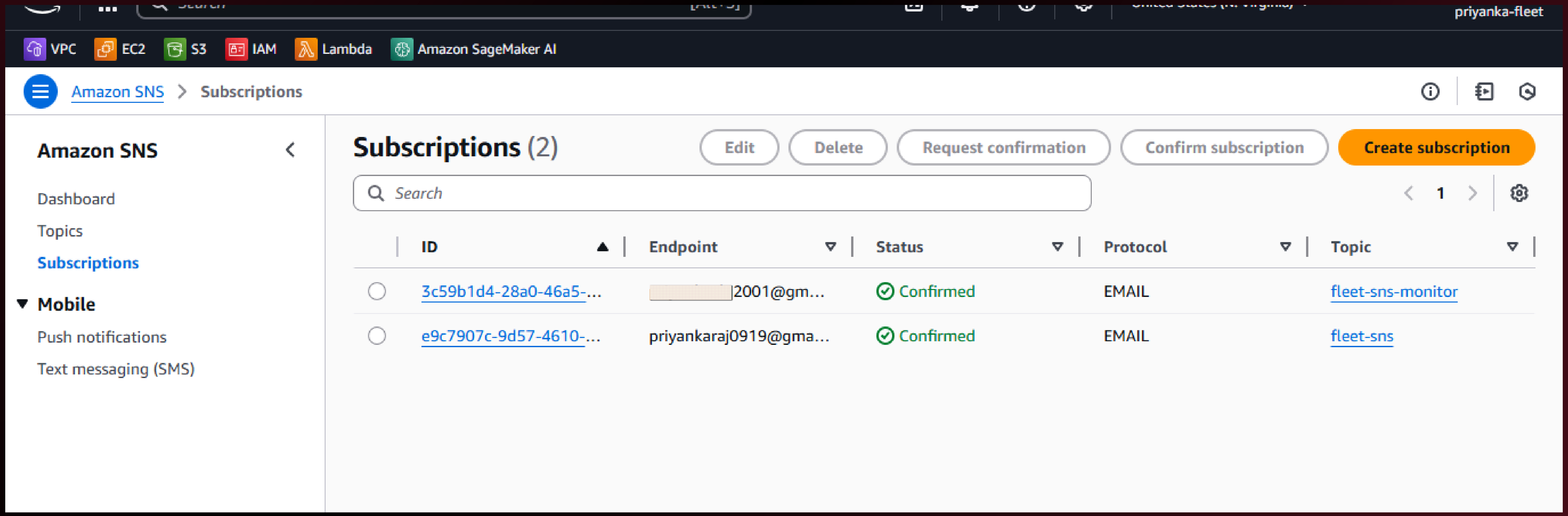




Created SNS topic,added mail address under subscription to receive fleet alert mail.

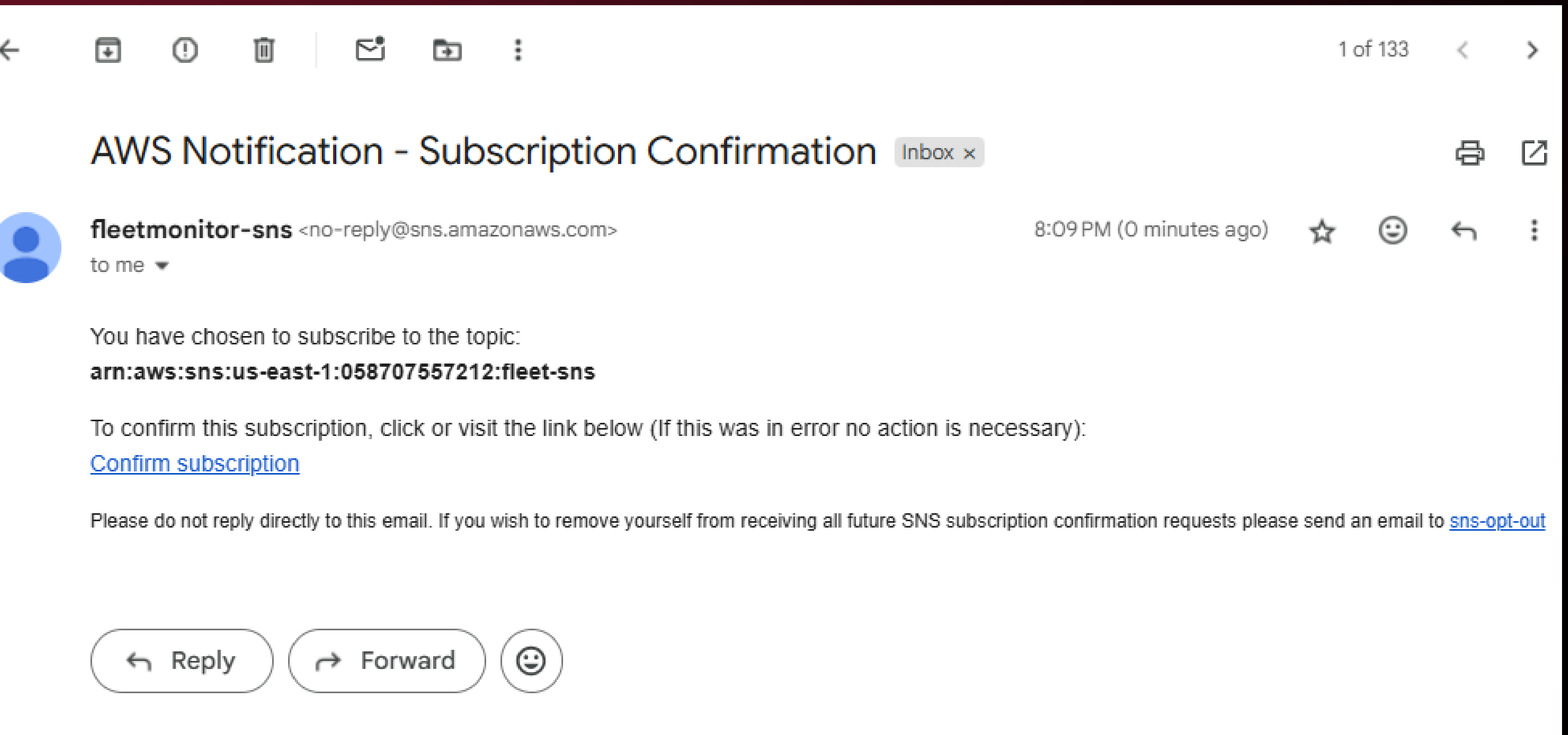
AMAZON SNS – TOPICS AND SUBSCRIPTIONS

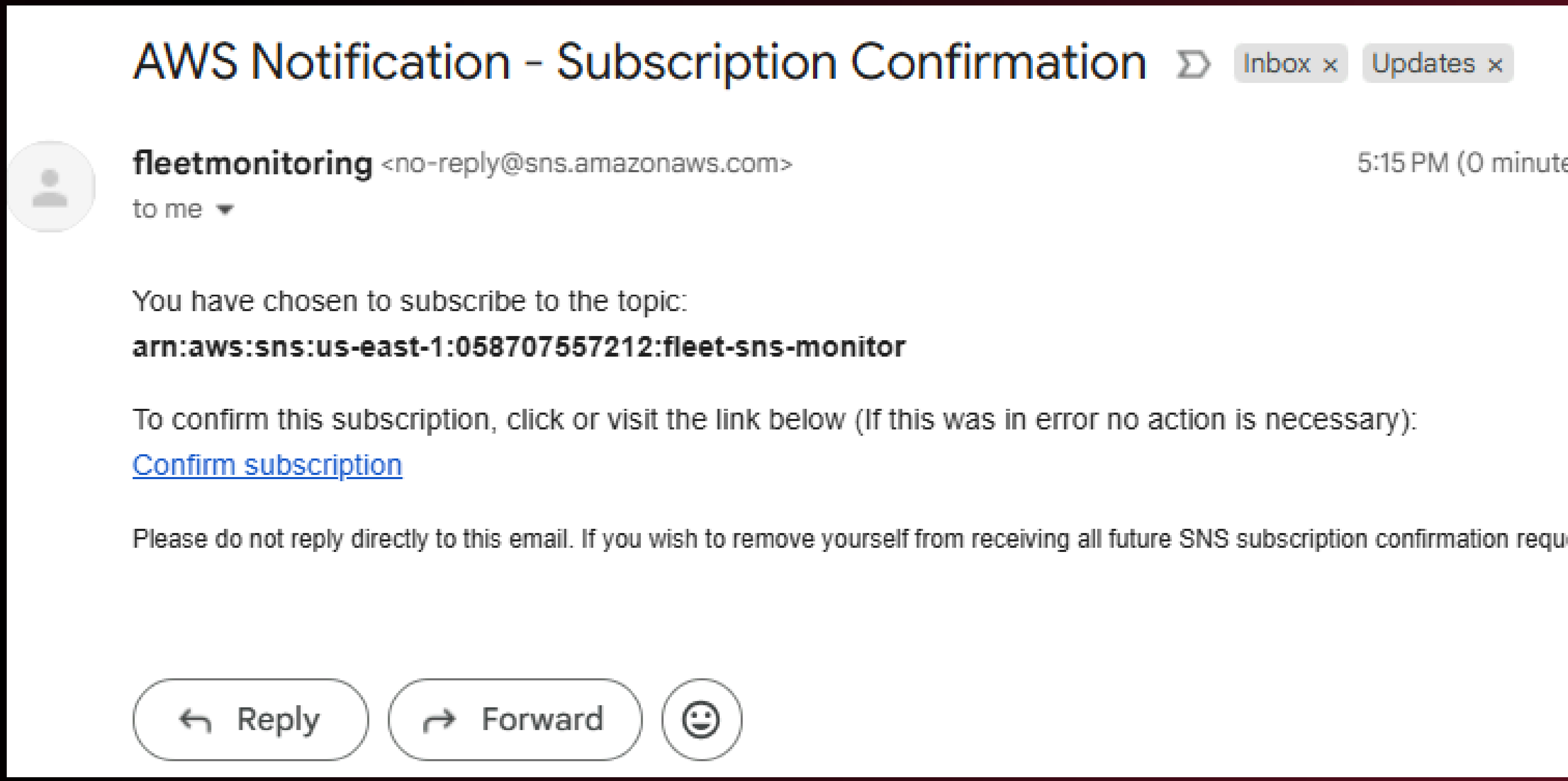


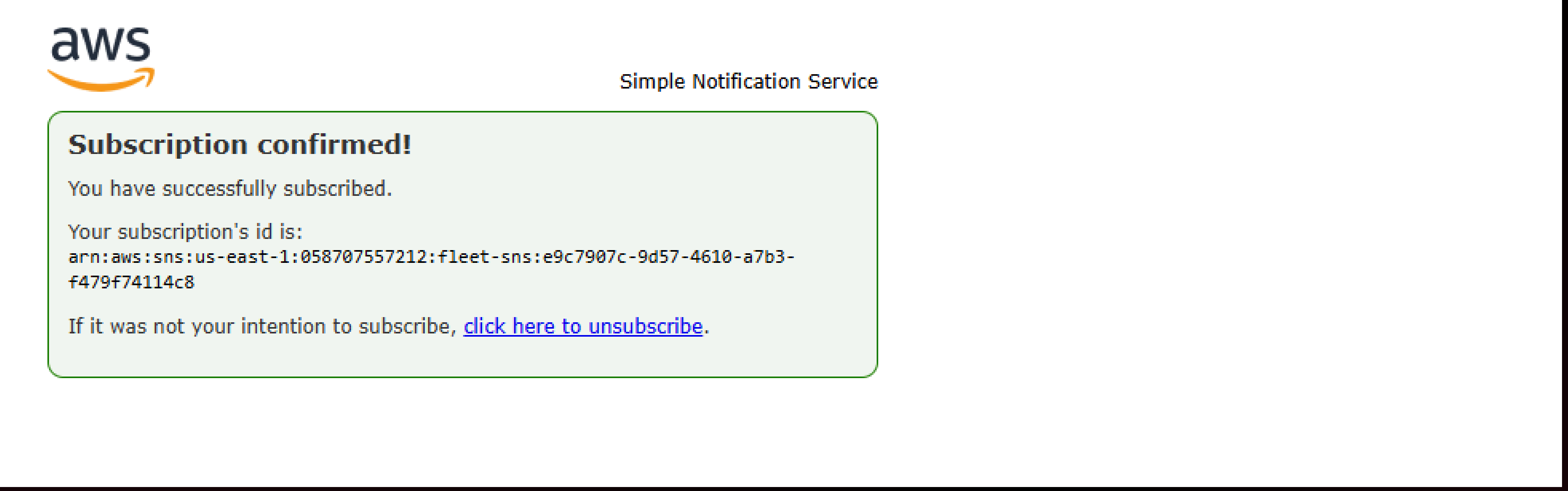


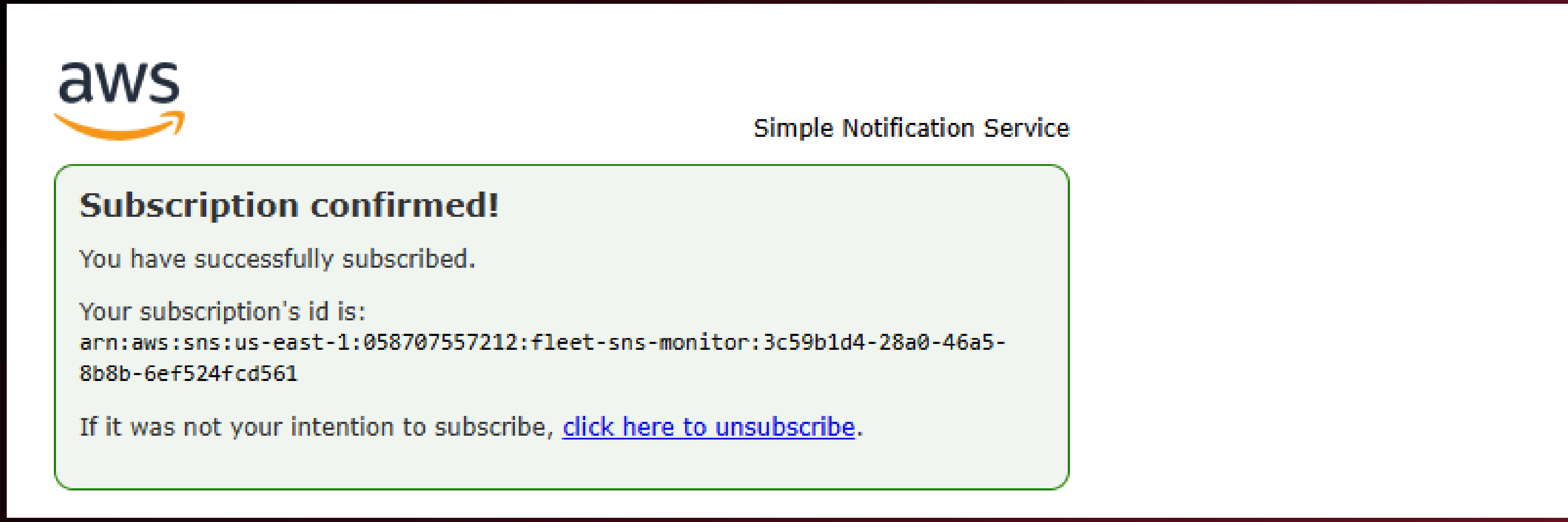
Added 2 topics and 2 Subscription mail address to check the demo.

CONFIRMATION MAIL



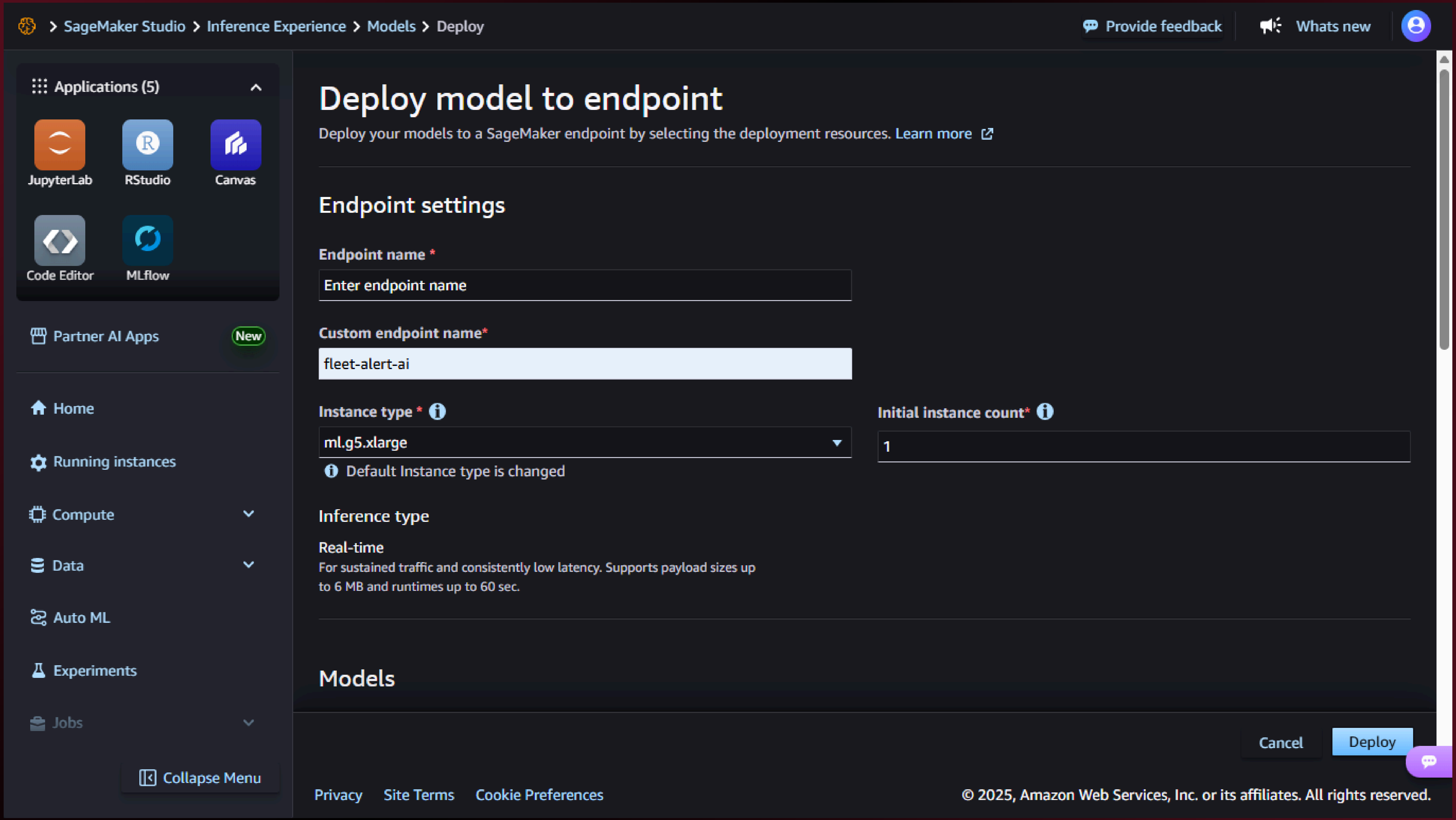






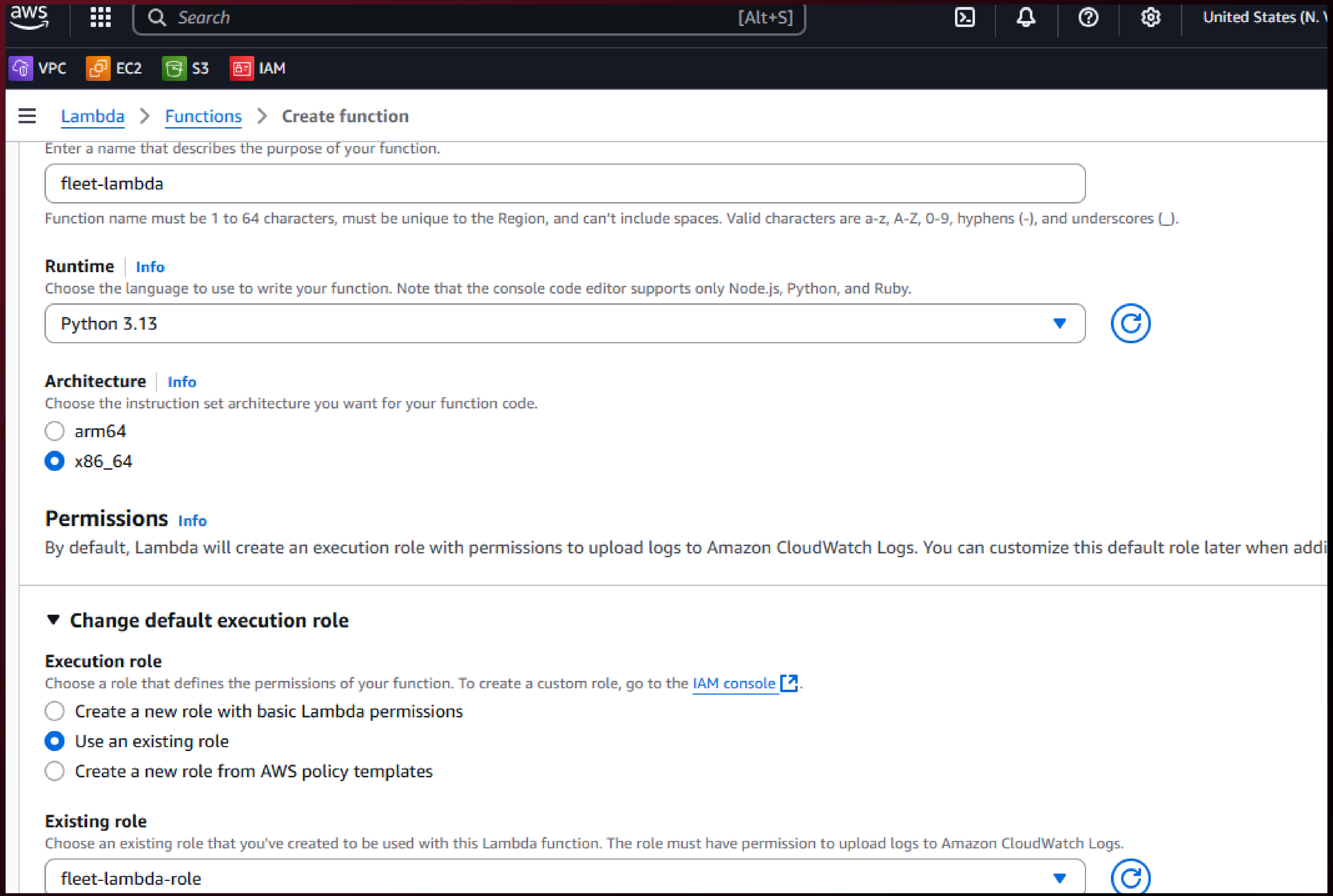
An Automated email for subscription is received, confirmed subscription to get the alert mails.

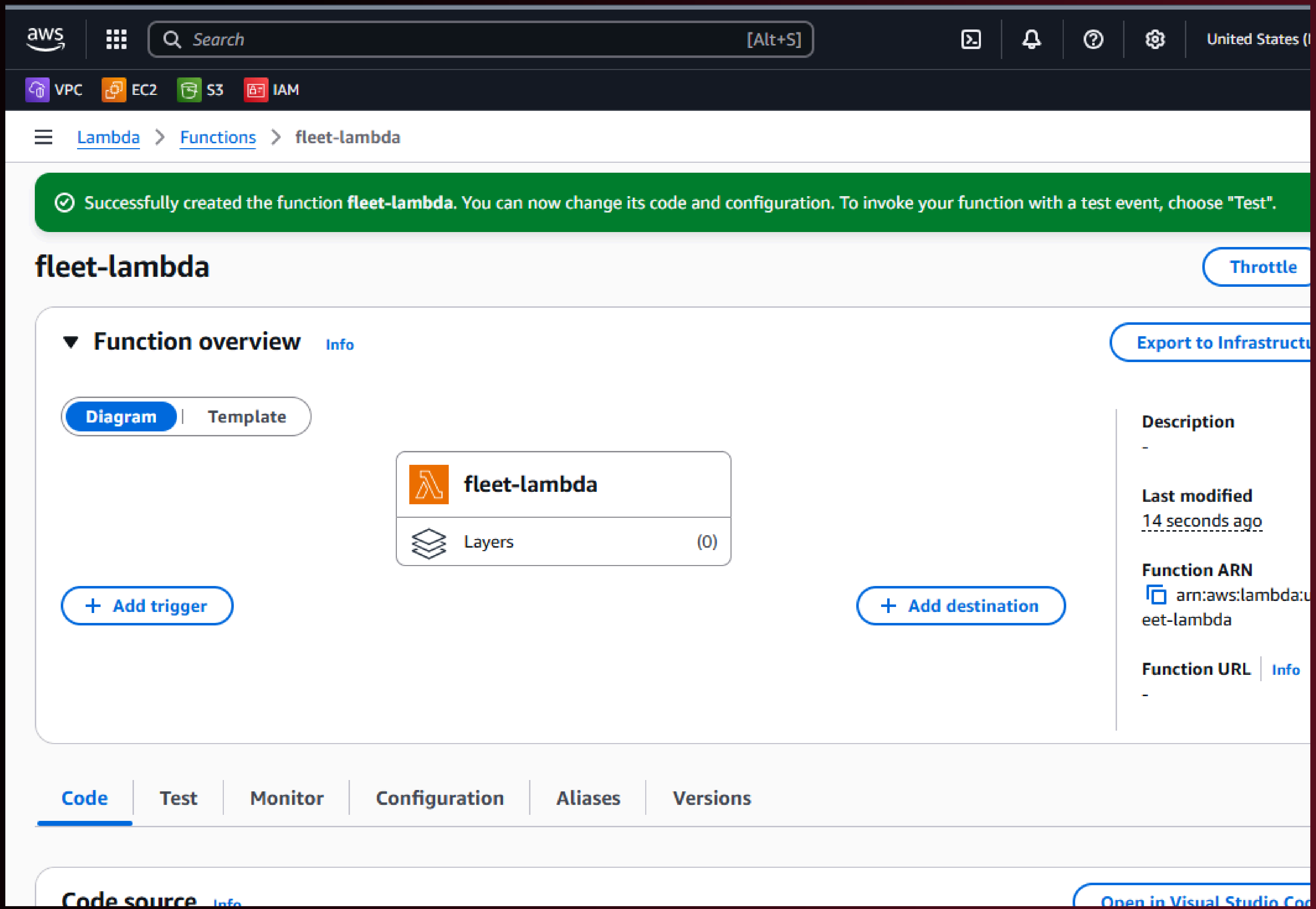
SAGEMAKER ENDPOINT



Creating SageMaker endpoint to be invoked by Lambda using a simple ML classification / regression model for anomaly detection on temperature data

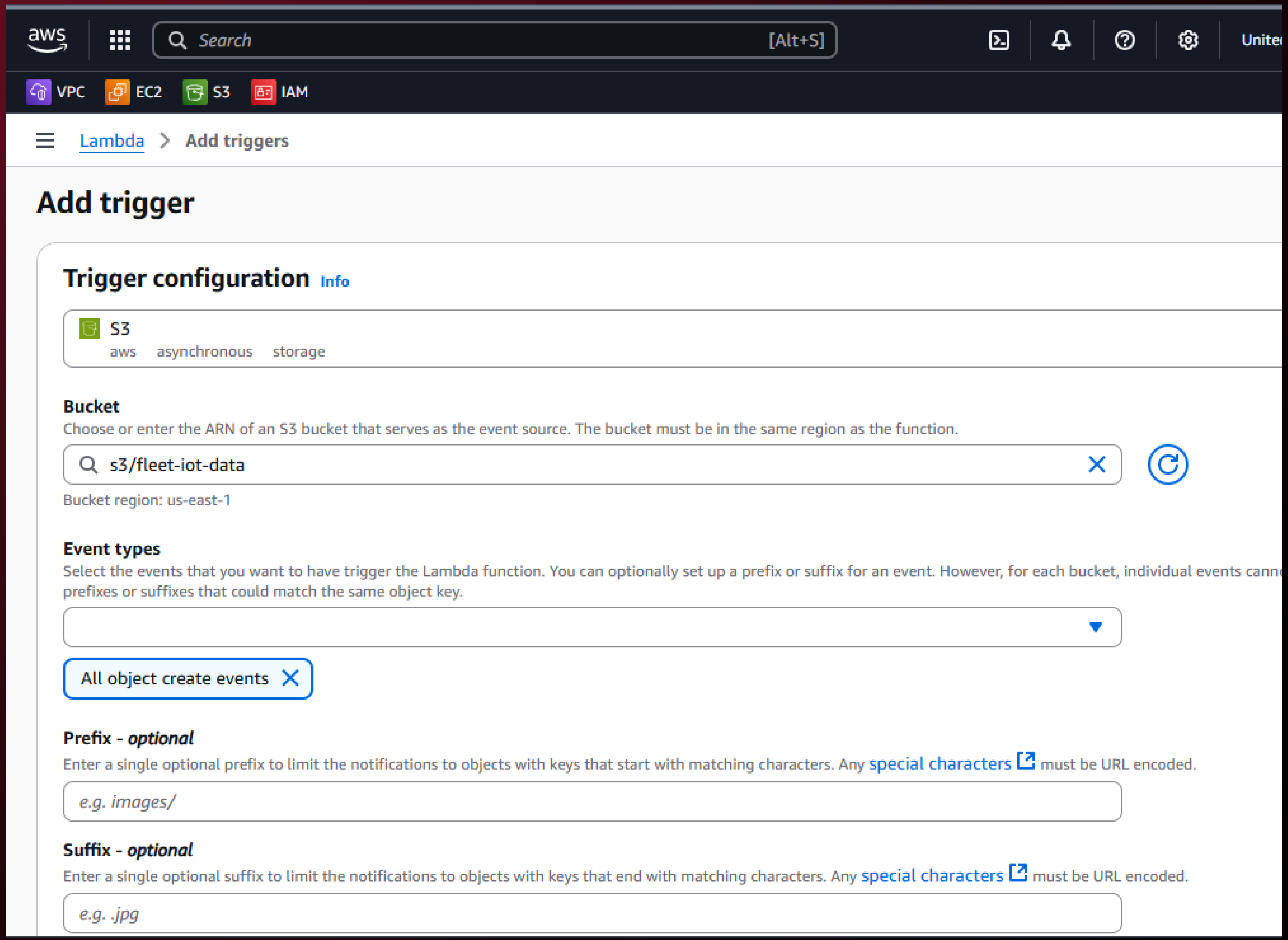
AWS LAMBDA

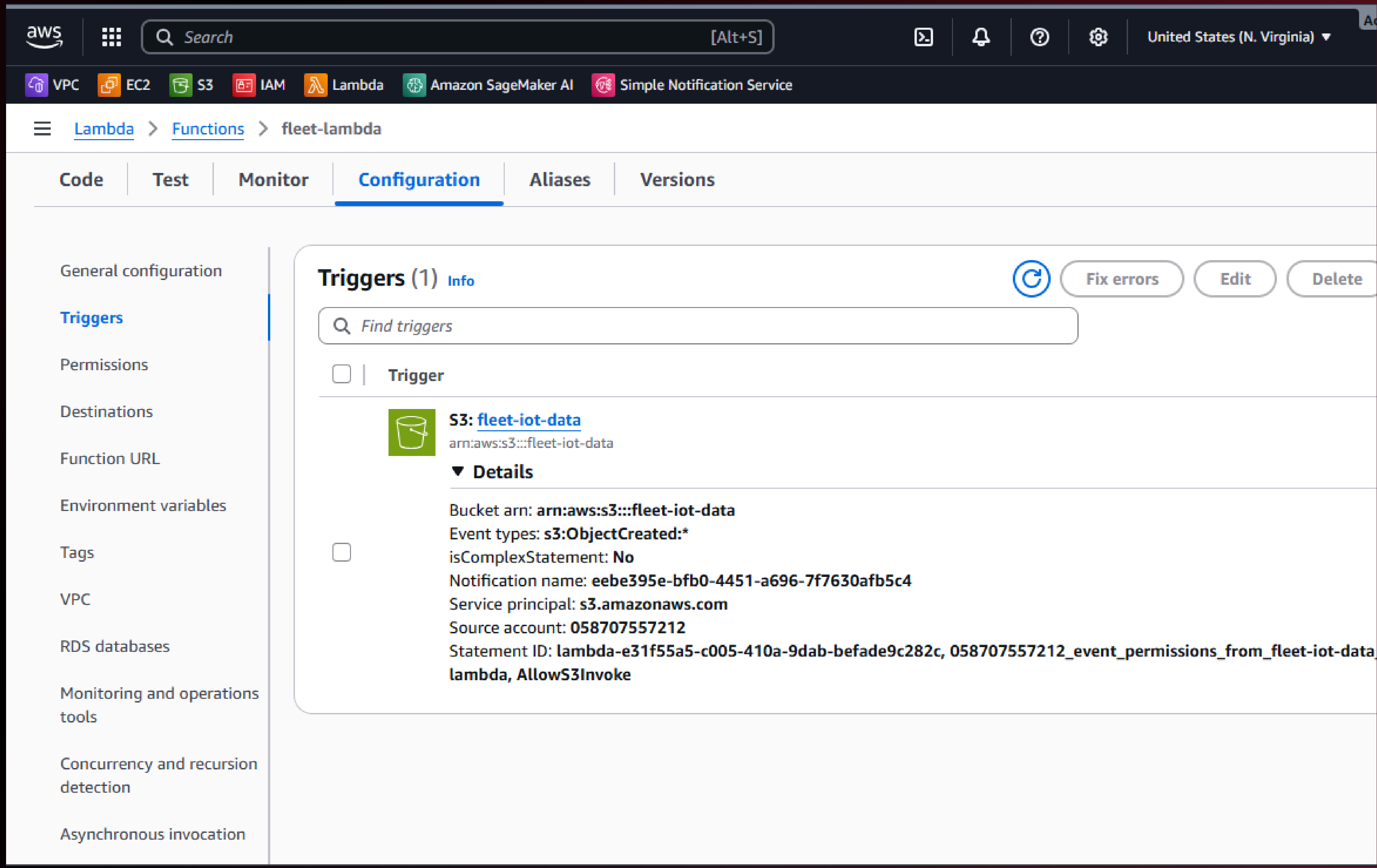




Created Lambda function with Runtime Python 3.13 and attached the respective role created.

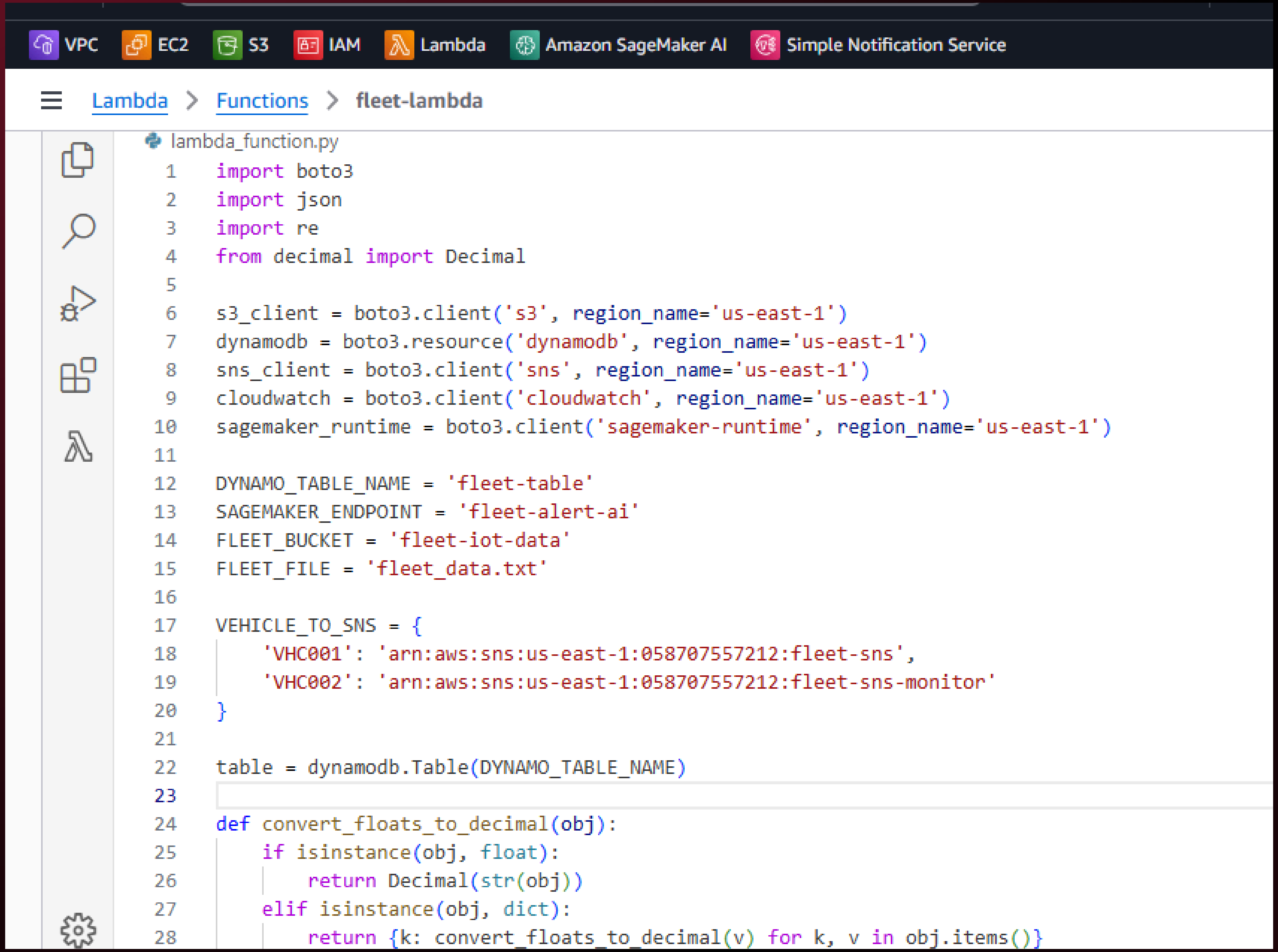
S3 TRIGGER FOR LAMBDA

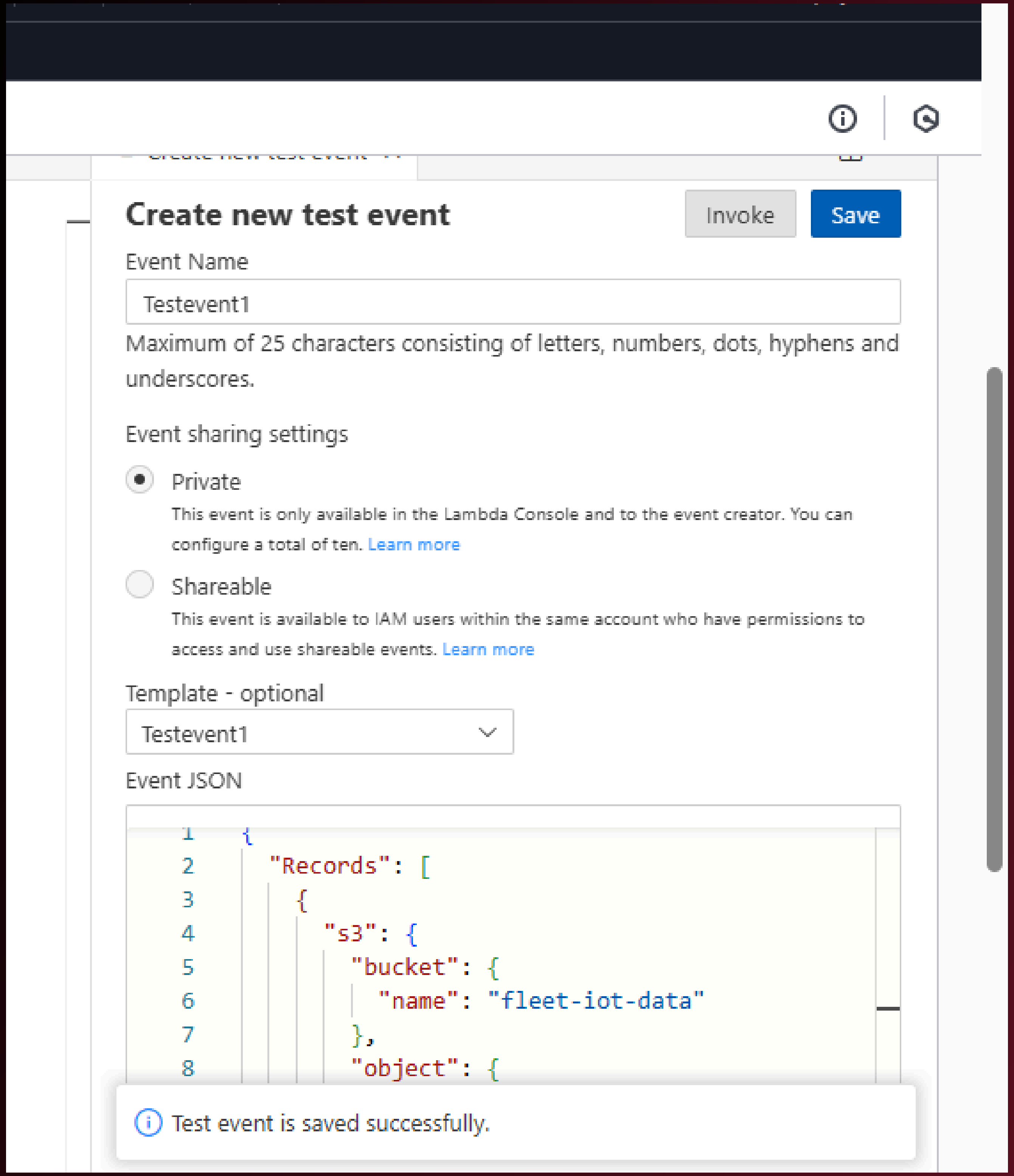




Added S3 trigger to lambda,with the required event and permissions.

LAMBDA FUNCTION





Lambda orchestrates the complete workflow using python ,ingesting data from S3, sending alerts, invoking SageMaker for ML predictions, and storing results in DynamoDB .For full code - Github

RAW DATA

fleet_data.txt

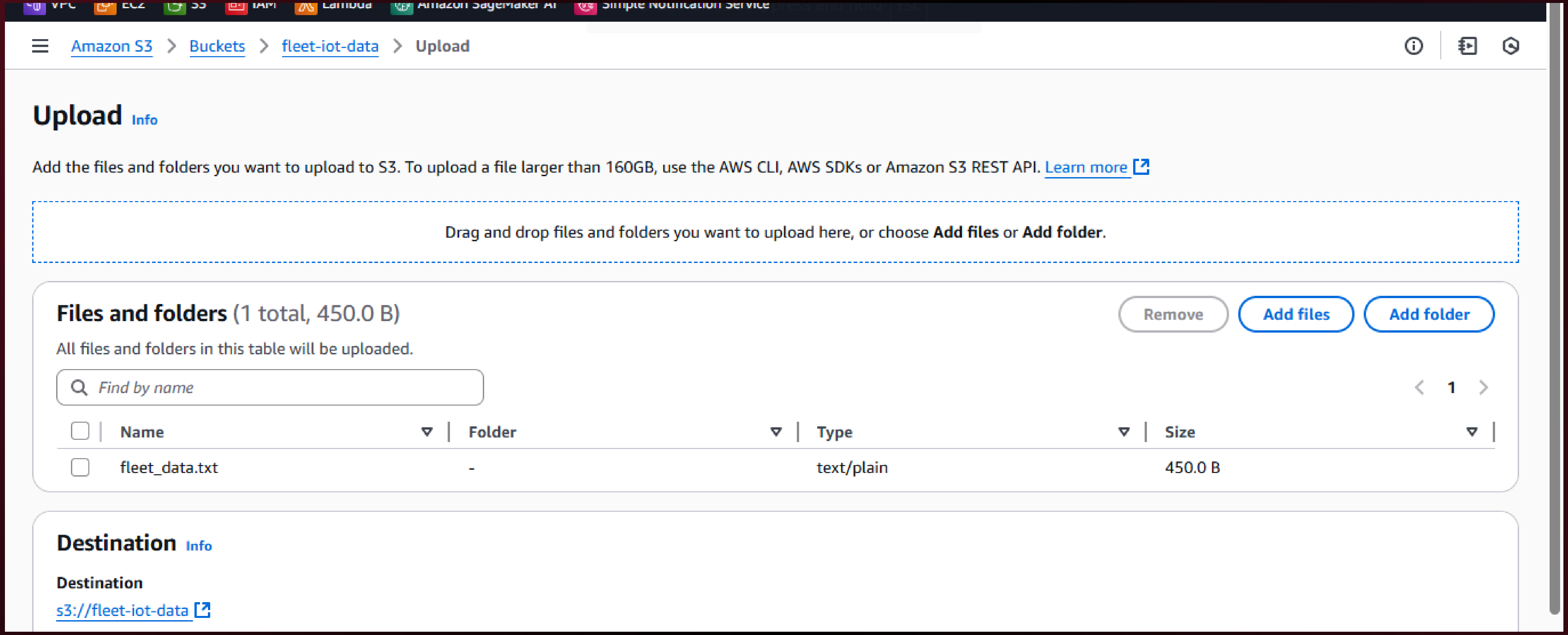
File Edit View

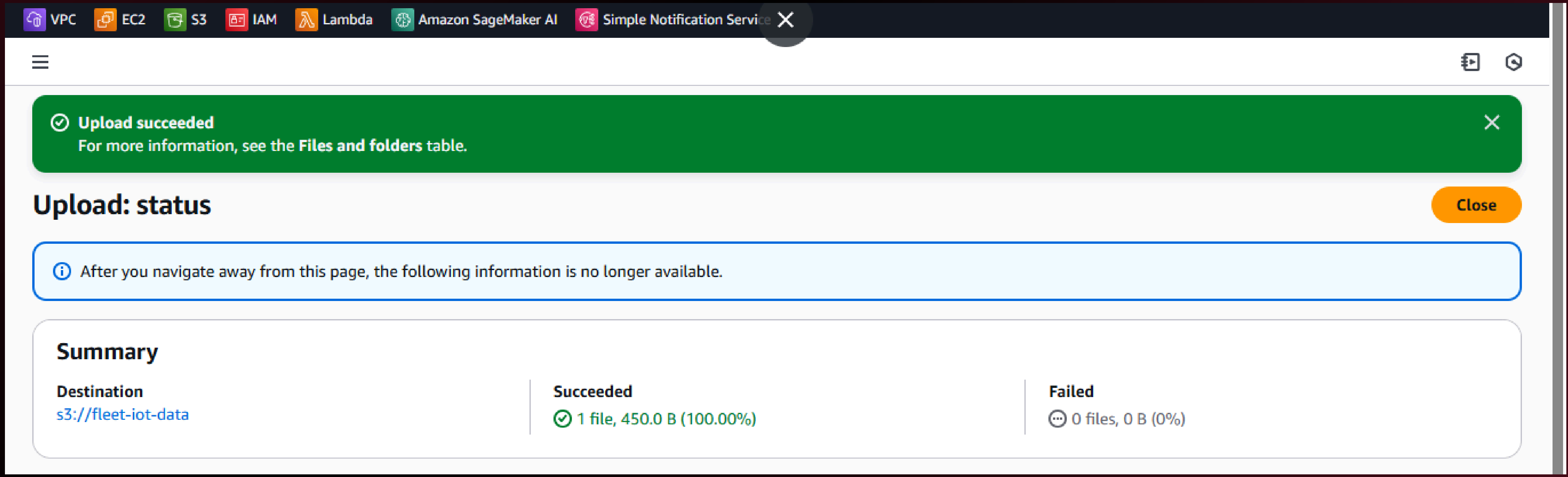
device_id VHC001 reported at timestamp 19-10-2025 22:38 with speed_kmph 73, fuel_level_percent 98, engine_temp_c 102, gps_latitude 12.398542, gps_longitude 79.921675, odometer_km 10365, and status maintenance_required.

device_id VHC002 reported at timestamp 19-10-2025 22:38 with speed_kmph 83, fuel_level_percent 98, engine_temp_c 124, gps_latitude 12.398542, gps_longitude 79.921675, odometer_km 10365, and status maintenance_required.

Using 2 raw sentence datas for demo.

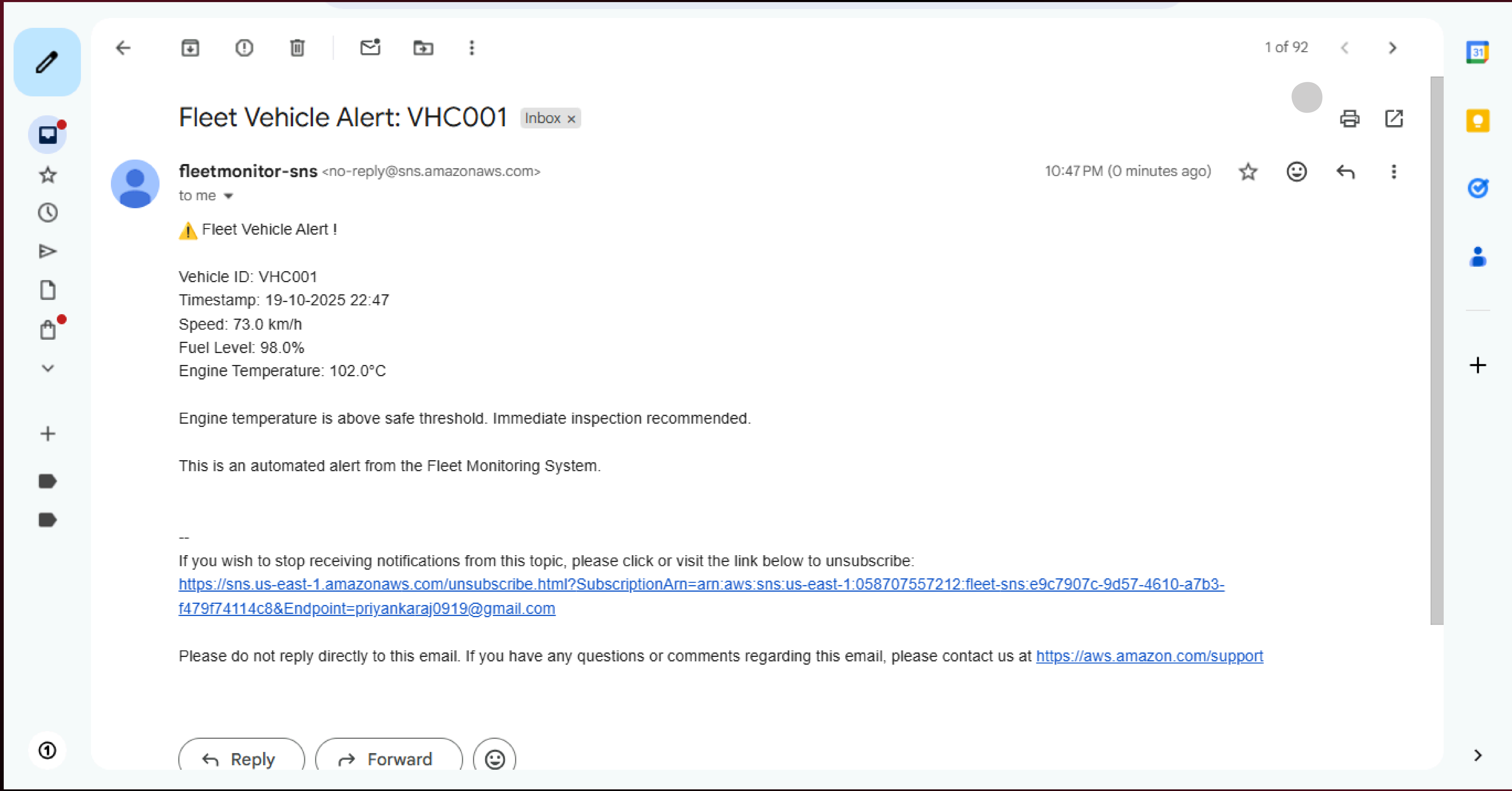
WORKING DEMO





Uploading the raw data file to s3 bucket.

FLEET ALERT MAIL


Fleet Vehicle Alert: VHC001 Inbox ×

1 of 92

fleetmonitor-sns <no-reply@sns.amazonaws.com>
to me

10:47 PM (0 minutes ago)

⚠️ Fleet Vehicle Alert !

Vehicle ID: VHC001
Timestamp: 19-10-2025 22:47
Speed: 73.0 km/h
Fuel Level: 98.0%
Engine Temperature: 102.0°C

Engine temperature is above safe threshold. Immediate inspection recommended.

This is an automated alert from the Fleet Monitoring System.

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If you wish to stop receiving notifications from this topic, please click or visit the link below to unsubscribe:
https://sns.us-east-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:us-east-1:058707557212:fleet-sns:e9c7907c-9d57-4610-a7b3-f479f74114c8&Endpoint=priyankaraj0919@gmail.com

Please do not reply directly to this email. If you have any questions or comments regarding this email, please contact us at https://aws.amazon.com/support

Reply Forward

Received the alert mail to the 1st Subscription mail id as the temperature is >100 degree.

1 of 77

Fleet Vehicle Alert: VHC002

Inbox ×

fleetmonitoring <no-reply@sns.amazonaws.com>

to me

10:47 PM (0 minutes ago)

⚠️ Fleet Vehicle Alert !

Vehicle ID: VHC002
Timestamp: 19-10-2025 22:47
Speed: 83.0 km/h
Fuel Level: 98.0%
Engine Temperature: 124.0°C

Engine temperature is above safe threshold. Immediate inspection recommended.

This is an automated alert from the Fleet Monitoring System.

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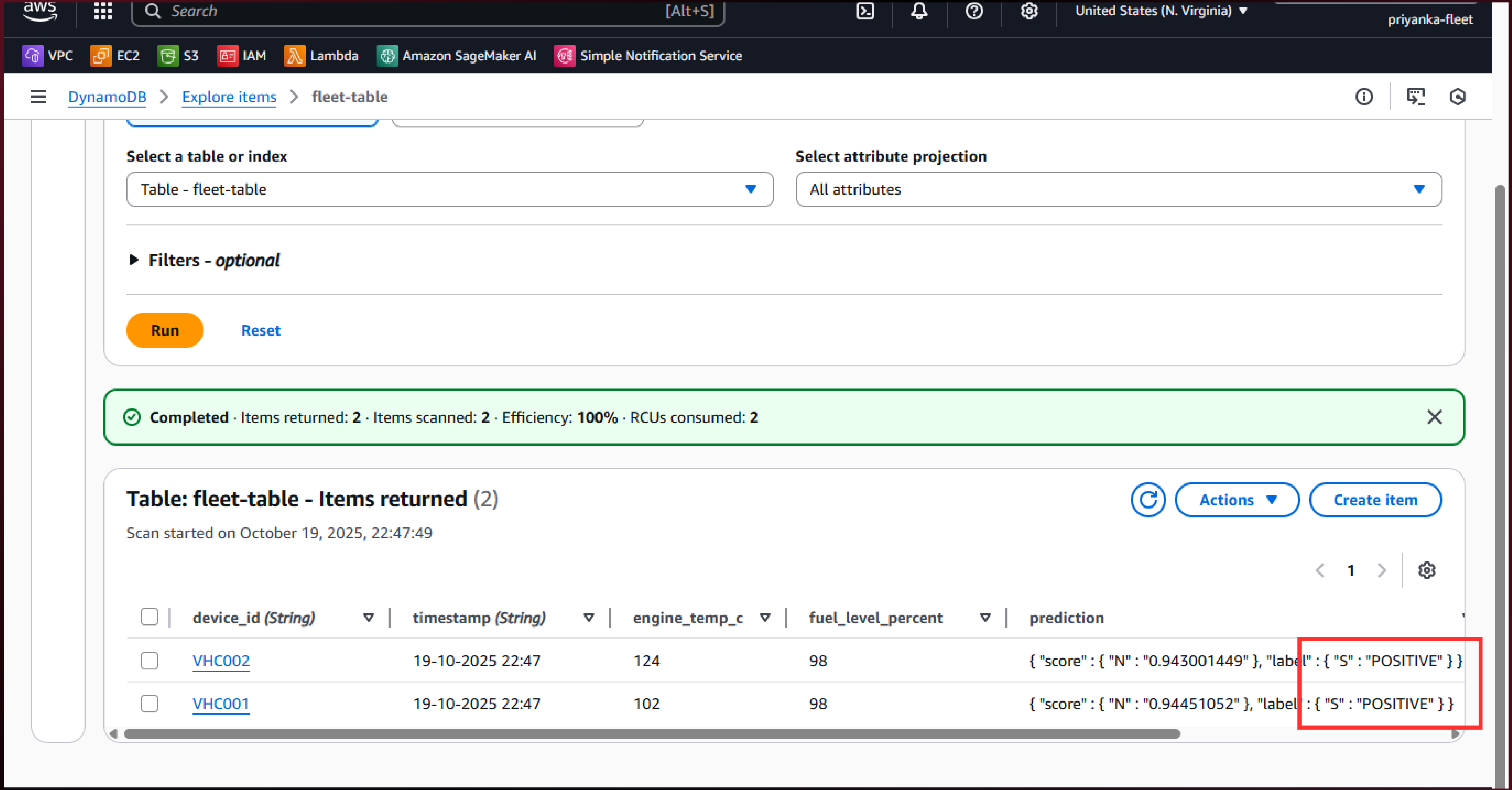
If you wish to stop receiving notifications from this topic, please click or visit the link below to unsubscribe:
https://sns.us-east-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:us-east-1:058707557212:fleet-sns-monitor:3c59b1d4-28a0-46a5-8b8b-6ef524fcd561&Endpoint=priyankaabi2001@gmail.com

Please do not reply directly to this email. If you have any questions or comments regarding this email, please contact us at https://aws.amazon.com/support

Reply Forward

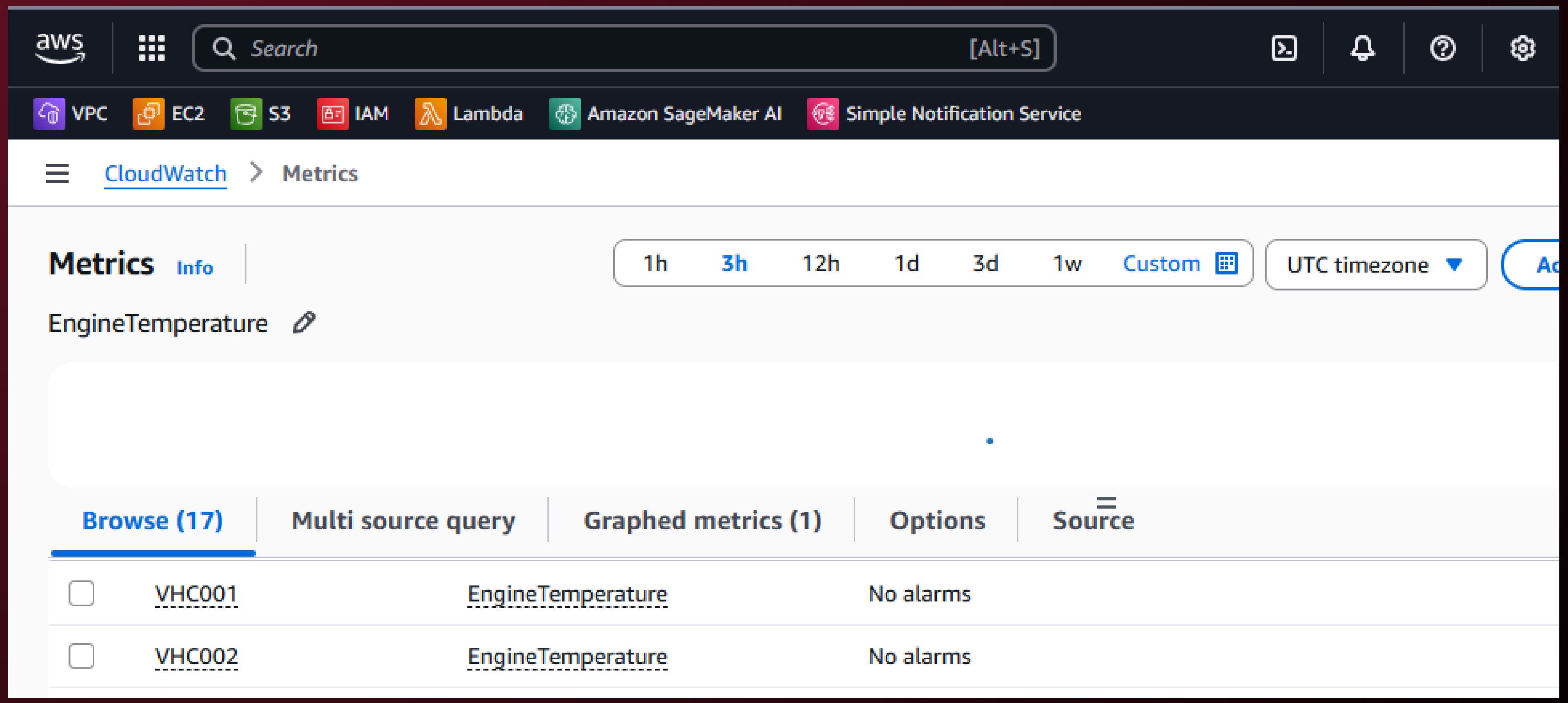
Received the alert mail to the 2st Subscription mail id added as the temperature is >100 degree.

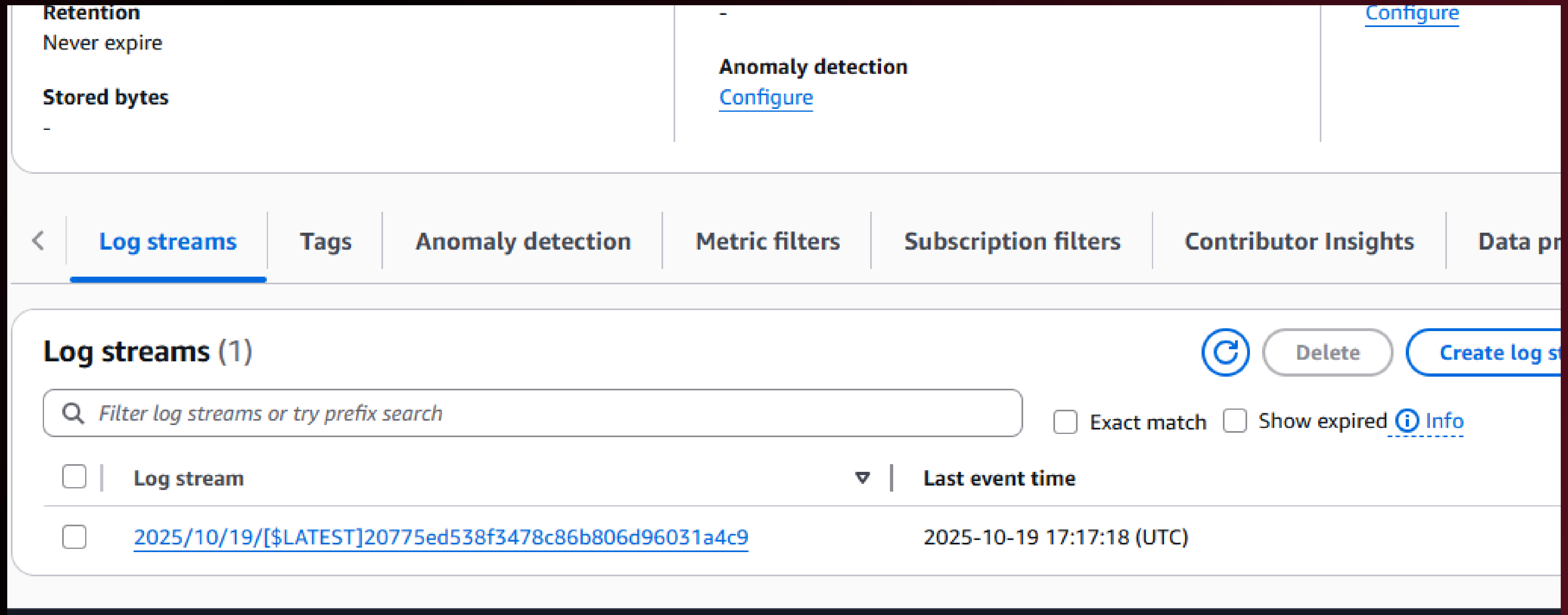
SAGEMAKER PREDICTION

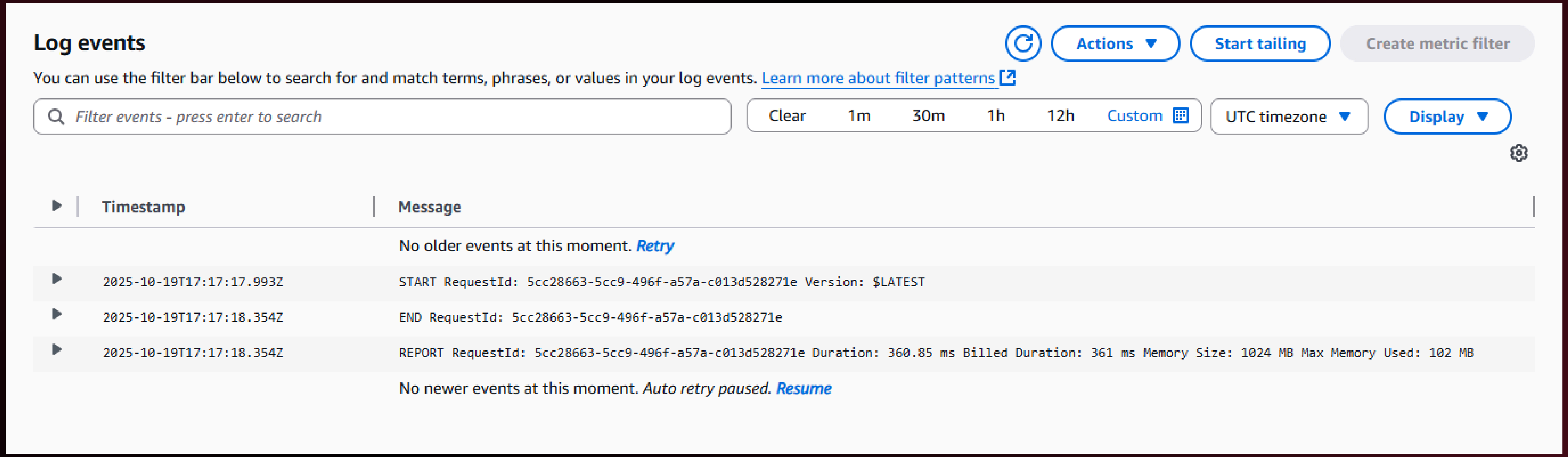


After uploading raw files in S3,Dynamo DB makes it a clear organised file and Sagemaker checks on S3 and predicts "POSITIVE".

CLOUD WATCH LOGS







Cloud watch shows the logs and metrics of the events done recently.

PROJECT SUMMARY

- Designed and implemented an end-to-end serverless fleet monitoring predictive maintenance system using AWS.
- Built an automated workflow across S3, Lambda, SageMaker, DynamoDB, and SNS.
- Achieved real-time data processing, prediction, alerting, and secure storage.
- Integrated IAM and KMS to ensure secure access and data protection.
- The system is scalable, cost-effective, and suitable for large fleet operations.
- Full code and detailed documentation are available in GitHub-Repository

THANK YOU

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